“Open label multi-center dose-escalation phase 1 Clinical Trial Assessing safety, tolerability and efficacy of the combination of Actinomycin D and venetoclax in Relapsed/Refractory (R/R) Acute Myeloid Leukemia (AML)”

Actinomycin D and Venetoclax for relapse/refractory AML

INTERVENTIONAL RESEARCH PROTOCOL INVOLVING HUMAN PARTICIPANTS CONCERNING A MEDICINAL PRODUCT FOR HUMAN USE

Version N°1.0 dated 03/04/2026
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Coordinating Investigator	**Dr Sofiane FODIL** Department of Hematology Saint-Louis Hospital, Paris Tel. +33 1 42 49 96 43 Email: sofiane.fodil@aphp.fr
Scientific Director	**Professor Raphaël ITZYKSON** Department of Hematology Saint-Louis Hospital, Paris Tel. +33 1 42 49 96 43 Email : raphael.itzykson@aphp.fr
Sponsor	**Petra SLIVOVA** Assistance Publique – Hôpitaux de Paris (AP-HP) Direction de la Recherche Clinique et de l’Innovation - DRCI (Clinical Research and Innovation Department) 1, avenue Claude Vellefaux - 75010 Paris Tel: +33 1 44 84 17 35 Mail: petra.slivova@aphp.fr
Coordinating Biostatistician	**Professor Lucie Biard** Institute Saint-Louis for Research, University of Paris Hôpital Saint-Louis 1, avenue Claude Vellefaux - 75010 Paris Tel: +33 1 42 49 97 42 Mail: lucie.biard@u-paris.fr
Entity responsible for monitoring the study	**Laure Gilles** URC St Louis (Clinical Research Unit) Hôpital Saint-Louis 1, avenue Claude Vellefaux – 75010 Paris Tel : +33 1 42 49 49 60
Safety monitoring	**Valentin Hosansky** Hôpital Saint-Louis 1, avenue Claude Vellefaux - 75010 Paris Tel : +33 1 40 27 42 68

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	Mail: valentin.hosansky@aphp.fr
Pharmaceutical coordination	**Robin Charreteur** 7, rue du Fer à Moulin – 75005 Paris Tel : +33 1 46 69 92 66 Mail: robin.charreteur@aphp.fr

Direction de la Recherche Clinique et de l'Innovation - DRCI (Clinical Research and Innovation Department)
Hôpital Saint Louis 75010 PARIS

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SIGNATURE page for a research PROTOCOL

Project Code: APHP251246 / EU Clinical Trial No: 2025-524728-22-01

Title: Open label multi-center dose-escalation phase 1 Clinical Trial Assessing safety, tolerability and efficacy of the combination of Actinomycin D and venetoclax in Relapsed/Refractory (R/R) Acute Myeloid Leukemia (AML)

Version no. 1.0 dated 03/04/2026

The study will be carried out in accordance with the protocol, with current good practices and with statutory and regulatory requirements.

Coordinating Investigator:
Dr Sofiane Fodil
Department of Hematology
Hospital Saint-Louis
1 avenue Claude Vellefaux
75010 Paris

Date:
Signature:

Sponsor
Assistance Publique – Hôpitaux de Paris
Délégation à la Recherche Clinique et à l'Innovation - DRCI (Clinical Research and Innovation Department)
Hôpital Saint Louis
1 avenue Claude Vellefaux
75010 PARIS

Date:
Signature:

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1 SUMMARY

Full title	Open label multi-center dose-escalation phase 1 Clinical Trial Assessing safety, tolerability and efficacy of the combination of Actinomycin D and venetoclax in Relapsed/Refractory (R/R) Acute Myeloid Leukemia (AML)
Acronym/reference	ACTIVATE
Coordinating investigator	Dr Sofiane FODIL
Scientific Director	Prof Raphaël Itzykson
Sponsor	Assistance Publique – Hôpitaux de Paris
Scientific justification	R/R AML is an unmet medical need. Acute myeloid leukemia (AML) is a heterogeneous clonal myeloid neoplasm where abnormal proliferation and impaired differentiation of hematopoietic stem and myeloid progenitor cells impede normal hematopoiesis. Despite the significant progress made in understanding AML oncogenesis over the past decades, this disease remains devastating with about 18,000 new cases every year in Europe and a five-year overall survival of only 17%[1]. The standard of care (SOC) of newly diagnosed AML includes a '7+3' cytarabine-anthracyclin combination followed by repeated courses of high dose cytarabine with or without allogeneic hematopoietic stem cell transplantation (HSCT)[2]. Up to one third of unselected AMLs are refractory to intensive chemotherapy. Outcome of such primary refractory AMLs is dismal, with a median survival shorter than one year[3]. Half of patients achieving a remission after intensive therapy eventually relapse. The majority of relapsed AMLs lack tractable FLT3 or IDH1-2 driver mutations and their treatment rely on intensive salvage therapy, with a minority of patients proceeding to a potentially curative HSCT[4]. Patients with newly diagnosed AML who are too old or frail for intensive chemotherapy receive less intensive therapies based on hypomethylating agents (HMAs), which, as single agents, are not curative[5]. Though the prognosis of such patients may improve in the coming decade with the addition of venetoclax to frontline HMAs[6], patients relapsing or refractory to HMA-based therapies also have survival rates lower than 12 months[7]. As such, R/R AML represents an unmet medical need. Novel treatment options independent of the conventional chemotherapy and HMA backbones used in the frontline setting are thus necessary. Actinomycin D and Venetoclax show synergism in AML. In 2018, venetoclax received accelerated approval by the FDA for use in AML patients. As a BH3 mimetic, venetoclax is a selective and oral inhibitor that binds to BCL-2, an important regulator of the apoptosis pathway by tightly controlling the release of cytochrome C from the mitochondria, the initiating step during apoptosis[8]. Actinomycin D is known to inhibit RNA Polymerase I (RNA Pol I), depleting nucleoli of ribosomal RNA (rRNA) and inducing nucleolar stress response in AML cells[9]. Actinomycin D has also been shown to repress MCL-1 expression and synergize with BH3 mimetics in cancer cell lines[10]. MCL-1 upregulation is a major venetoclax

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	resistance pathway[11]. Lately, it has been shown that actinomycin D targets primed mitochondria in AML cell driving TP53 activation and senescence of AML cells. Based on the pharmacodynamics of these two drugs, pre-clinical data have demonstrated the synergistic effect of venetoclax and actinomycin D[12]. Actinomycin D as a single agent has been used in R/R AML and retrospective data published showed that four out of the nine evaluable patients achieved CR/CRi[9]; or that three out of 17 patients with morphologic relapses or refractory AML achieved complete morphologic remission (CR) after the first cycle of Actinomycin D[13]. Retrospective data about the actinomycin D-venetoclax combination in R/R AML are available with overall response rate ranges from 26%[14] to 75%[15]. Interestingly, responses to the combination of ActD-Ven were also noted in patients previously exposed to venetoclax. Finally, no treatment-related death was reported in those retrospective series[14,15]. Collectively, these findings provide a strong rationale for the clinical evaluation of Actinomycine D-Venetoclax combination in R/R AML patients. The purpose of this phase 1 study is to evaluate the safety and feasibility of such strategy, provide preliminary signals of efficacy, and identify potential biomarkers.
Main objective and primary endpoint	<u>Main objective:</u> To identify the maximum tolerated dose (MTD) of Actinomycin D / Venetoclax combination in patients with R/R AML. <u>Primary endpoint:</u> Dose Limiting Toxicity (DLT) DLT will be defined during a safety observation period corresponding to the first cycle of Actinomycin D/Venetoclax treatment, as: ● Death of any cause (with the exception of deaths caused by clear evidence of disease progression, defined as a leukocytosis greater than 50 × 10⁹/L) ● Prolonged myelosuppression on day 42 from start of therapy or later without evidence of leukemia defined by the absence of platelet recovery above 50 × 10⁹/L without transfusion support and the absence of neutrophil recovery with ANC > 1 × 10⁹/L ● Grade ≥ 3 non- hematological toxicity with the exception of: o Grade 3 or 4 infections o Grade 3 or 4 fever with neutropenia o Grade 3 nausea, vomiting or diarrhea that can be managed to ≤ Grade 2 within 72 hours of symptomatic treatment o Grade 3 mucositis or stomatitis that can be managed to ≤ Grade 2 within 72 hours of symptomatic treatment

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	○ Grade 3 tumor lysis syndrome that resolves within 72 hours of onset with medical treatment ○ Grade 3 or 4 hemorrhage Probability of DLT should not exceed 33% at the end of the first cycle of Actinomycin D/Venetoclax (Day 28-42).
Secondary objectives and endpoints	<u>Secondary objectives:</u> 1) To characterize safety of the combination Actinomycin D/ Venetoclax 2) To preliminarily assess the anti-leukemia efficacy of the combination of Actinomycin D/ Venetoclax treatment. 3) To assess the combined pharmacokinetics and pharmacodynamics of Actinomycin D/ Venetoclax treatment in leukemia cells of peripheral blood and bone marrow. <u>Secondary endpoints:</u> *1) Safety evaluation* ● AE, TEAE, treatment related TEAE, TEAES leading to treatment discontinuation, serious adverse events (SAE): Frequency, duration and severity ● Safety outcome measures will be assessed continuously during the study. ● Monitoring of ECGs and clinical laboratory values are integral to safety assessment. ● Adverse events (AE), treatment emergent adverse events (TEAE) and treatment-related TEAE will be evaluated according to the NCI CTCAE version 6.0. Serious adverse events will be defined per protocol according to EMA/ICH criteria. Causality relationship with the experimental treatment should be assessed by the investigator. ● Patients should be followed until all treatment related adverse event have returned to grade ≤ 1 or baseline or are deemed irreversible by the investigator. *2) Preliminary assessment of anti-leukemia efficacy* ● **Complete response rate** including CR, CRi (CR with incomplete hematologic recovery, meaning CR with platelet count <100,000/µL or absolute neutrophil count <1000/µL) and CRh (CR with partial hematologic recovery, meaning CR not fulfilling CR or CRi peripheral blood count criteria but with platelet count >50,000/µL AND absolute neutrophil count >500/µL). ● **Morphologic leukemia-free state** (MLFS): blastic criteria of response on the bone marrow aspirate, absence of extramedullary disease and no requirement of hematologic recovery.

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	● **Partial response** (PR): absence of circulating blasts; absence of extramedullary disease; decrease of bone marrow blast percentage from 5 to 25% and decrease of pretreatment bone marrow blast percentage by at least 50% ● **Overall response rate** (ORR) being defined as CR, CRi and CRh, MLFS and PR rates. ● **Event-free survival** (EFS); here events are defined as: o Non achievement of hematologic response including CR, CRi, CRh o Hematologic relapse or progressive disease o Initiation of any subsequent anti-leukemic therapy (excluding hydroxyurea) o Deaths prior to one of the events mentioned above Reasons for subsequent therapy initiation (including intolerance, resistance, hematologic relapse, unsatisfactory MRD evolution), its type (intensive, less-intensive, best supportive care) and its frame (clinical trial, off-label or compassionate use) will be prospectively recorded. ● **Progressive disease** (PD): o >50% increase in marrow blasts over baseline (a minimum 15 points increase is required in cases with <30% blasts at baseline); or a persistent marrow blast percentage of >70% over at least 3 months, (without at least a 100% improvement in ANC to an absolute level >0.5x10^9/L and/or platelet count to > 50 x 10^9/L, non-transfused); o Or a new extramedullary disease ● **Duration of response** (DOR), ● **Relapse free survival** (RFS) ● **Overall survival** (OS) ● **Early mortality rates after 1 cycle of treatment** (day 28-42) and **after 2 cycles** (day 56-84) ● **Incidence of subsequent allogeneic HSCT**, overall and in responding patients specifically ● **Molecular Residual Disease** (MRD) will be evaluated as follow: Peripheral blood RQ-PCR for NPM1 patients[16], BM RQ-PCR for t(8;21) and inv(16) patients[17], otherwise FCM. Apart from specific mentions, all these hematological endpoints will be defined according to the last recommendation of the European Leukemia Network (ELN; currently the 2022 ELN recommendations in *Döhner H et al Blood 2022*). *3) Pharmacokinetics/Pharmacodynamics* To assess the combined pharmacokinetics and pharmacodynamics of Actinomycin D/Venetoclax treatment in leukemia cells of peripheral blood and bone marrow, during the clinical trial, with research of biomarkers of response
Design of the study	Dose-escalation phase 1, open label, multicenter clinical trial following an adaptive Bayesian time to event continual reassessment

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	design (TITE-CRM), to identify the optimal schedule of Actinomycine D in combination with Venetoclax days 1-7. The design will be applied to a single, specific population: adult patients with a confirmed diagnosis of relapsed or refractory AML, regardless of prior treatments or number of relapses. Number of Samples: 36 patients in total
Population of study participants	Patients aged 18 years old or more with R/R AML.
Inclusion criteria	● Patients aged 18 years old or more ● Patients with confirmed diagnosis of R/R AML according to the 2022 ELN recommendations who have failed available standard therapies, are not eligible for standard therapeutic options, or for whom no appropriate standard treatment option is considered available according to the investigator's judgment ● WBC < 20,000/mm3 (previous HY is allowed to control WBC count) ● ECOG performance status ≤2 ● Any prior treatment-related toxicities resolved to ≤ Grade 1 prior to first study dose, with the exception of ≤ Grade 2 neuropathy or alopecia ● AST and ALT ≤3.0 x upper the limit of normal (ULN) and total and direct serum bilirubin ≤ 1.5 x ULN unless considered due to leukemia, in which case thresholds may be increased to ≤ 6.0 × ULN for transaminases and ≤ 3.0 × ULN for bilirubin. ● Estimated glomerular filtration rate (GFR) ≥ 50 mL/min/1,73m^2 according to the MDRD equation ● Posaconazole prophylaxis for invasive pulmonary aspergillosis (IPA) ● Written informed consent obtained prior to any screening procedures ● Patients affiliated or beneficiary of the National Health Insurance in France (AME excluded)
Exclusion criteria	− Acute promyelocytic leukemia − Proven central nervous system leukemic involvement − Prior allogeneic HSCT within the last 12 weeks or donor lymphocyte conditioning (DLI) within the last 4 weeks − Graft-Versus-Host Disease (GVHD): Signs or symptoms of acute GVHD ≥ Grade 2, moderate or severe limited chronic GVHD or extensive chronic GVHD of any severity within the last 4 weeks prior to the first study dose. Patients must have been off all systemic immunosuppressive therapy for at least 2 weeks prior to enrollment. Patients may receive as GVHD treatment corticosteroid doses ≤ 10 mg of prednisone daily. − Concurrent therapy with any cytotoxic drug within 3 weeks before the first study dose. Only hydroxyurea for the control of blood counts is permitted. − Adults weighting less than 40 kg − Uncontrolled systemic fungal, bacterial, or viral infection (defined as ongoing signs/symptoms related to the infection without improvement despite appropriate treatment)

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	- HBV or HCV hepatitis with a detectable viral load or any ongoing anti-viral treatment - HIV with a detectable viral load or any ongoing anti-viral treatment - Other uncontrolled or active malignant disease within prior 12 months (excluding myelodysplastic syndrome; myeloproliferative neoplasm, cutaneous basal cell carcinoma, “in-situ” carcinoma of the cervix or breast, or other local malignancy excised) - Inability to swallow - Known malabsorption syndrome or other condition that may significantly impair absorption of oral study medications - Administration of any therapy that is considered to be investigational (i.e., used for non-approved indications(s) or in the context of a research investigation) within 3 weeks or 5 drug half-lives (whichever is longer) prior to the first dose of study drug - Any concurrent severe and/or uncontrolled medical condition, which could compromise participation in the study - Adult subjects to a legal protection order or unable to give their consent (*tutelle, curatelle, sauvegarde de justice, habilitation familiale, mandate de protection future*) - Persons deprived of their freedom by judicial or administrative decision, person hospitalized without their consent by virtues of articles L 3212-1 and L3213-1 and who are not subject to the provisions of article L 1121-8 - Females who are pregnant or breastfeeding - For males whose sexual partner is a woman of childbearing potential: unwillingness or inability of the man or woman to use a highly effective contraceptive method for the entire treatment period and for at least 6 months after completion of protocol treatment - For heterosexually active woman of childbearing potential: unwillingness or inability to use a highly effective contraceptive method for the entire treatment period and for at least 6 months after the administration of the final dose of study medication - Known allergies to investigational drug or to Posaconazole and/or excipients - Concomitant administration of St. John's Wort or yellow fever vaccine. Highly effective contraception methods include: combined (estrogen and progestogen containing) or progestogen-only hormonal methods associated with inhibition of ovulation, intra-uterine device, surgical sterilization (including bilateral tubal occlusion, partner’s vasectomy), or sexual abstinence if this is the preferred and usual lifestyle of the patient. In accordance with the venetoclax SmPC, women using hormonal contraception must also use an additional barrier contraceptive method during treatment and for the applicable post-treatment contraception period Women are not regarded as of childbearing potential if they are post-menopausal (at least 2 years without menses) or are surgically sterile (at least 1 month before enrolment).

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	Female patients must not donate or retrieve, for their own use, ova from the time of screening and throughout the treatment period, and for 12 weeks after the administration of the final dose of study medication. Female patients must agree not to breastfeed from the time of screening and throughout the protocol period, and for (5 half-lives) days after the administration of the final dose of study medication. Male patients must not freeze or donate sperm starting at screening and throughout the treatment period and 3 months after the administration of the final dose of study medication.
caInvestigational medicinal product(s)	Treatment will be delivered in consecutive cycles of 28 days of administration. Cycle windows may be extended beyond D28 upon prolonged myelosuppression, for monitoring until recovery. Patients will receive for two cycles: - Venetoclax daily days 1-7, - And Actinomycin D by short IV infusion 12.5 µg/kg/d for up to 3 days starting on D3. Patients deriving clinical benefit (defined as achievement of MLFS, CRi, CRh or CR) after 2 cycles may receive up to 10 consolidation cycles outside of the study, at the investigator's discretion, consisting of: - Venetoclax daily days 1-7, - And Actinomycin D by short IV infusion 12.5 µg/kg/d at day 3. Patients with a clinical benefit eligible for allogeneic transplantation, at the investigator's discretion, may receive up to 6 cycles prior to transplantation, outside of the study. Dose Levels for Actinomycin D are based on protocol available with safety data[15] in keeping with FDA guidance for the co-development of two drugs[18]. Actinomycin D will start on d3, after initial ramp up dosage of venetoclax, up to 3 days. Venetoclax will be given with a ramp up dosage over 3 days (20-50-70mg) until the daily dose of 70mg/d in patient with azole antifungal prophylaxis starting at D1. Venetoclax is given until D7 in each cycle to avoid further hematological toxicity. Three dose schedules of actinomycin D will be explored in induction phase, starting at DL1: • DL1: Venetoclax daily days 1-7 and Actinomycin D 12.5 µg/kg/d at day 3, • DL2: Venetoclax daily days 1-7 and Actinomycin D 12.5 µg/kg/d at days 3-4, • DL3: Venetoclax daily days 1-7 and Actinomycin D 12.5 µg/kg/d at days 3-5,

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	Dose escalation will be conducted according to a dose finding design with sequential dose assignment with cohorts of size 3, the number of patients eventually assessed at each dose level depending on the DLT.
Comparator treatment	None (one-arm study)
Interventions added for the study	Interventions added by the studies are limited to the following procedures: – Actinomycin D administration in combination with Venetoclax administration – PD exams from peripheral blood samples (plasma, serum or cells) – Additional volumes of bone marrow aspirations
Expected benefits for the participants and for society	The ACTIVATE trial aims to provide new therapeutic options for patients with relapsed/refractory AML, a population with limited treatment alternatives and poor prognosis. Participants in the study may benefit from potentially improved response rates. The combination of Actinomycin D and Venetoclax has shown synergistic anti-leukemic effects, which could lead to higher remission rates and prolonged survival. Furthermore, identifying biomarkers of response may contribute to personalized medicine approaches, optimizing treatment strategies for future patients. On a broader scale, this trial could expand the therapeutic landscape for R/R AML, potentially leading to new standard-of-care treatment options and improved patient outcomes worldwide.
Risks and burdens added by the study	Participation in the ACTIVATE trial introduces potential risks and burdens associated with the investigational treatment. Actinomycin D and Venetoclax can cause hematologic toxicities, including prolonged myelosuppression, neutropenia, and thrombocytopenia, which may increase the risk of infections and bleeding complications. Non-hematologic toxicities of Actinomycine D such as gastrointestinal disturbances (nausea, diarrhea, mucositis) are also expected. Patients will require frequent hospital visits and hospitalization, laboratory assessments, bone marrow aspiration increasing the overall burden. The potential for tumor lysis syndrome (TLS) necessitates strict hydration protocols and prophylactic measures. Despite these risks, the study is designed to closely monitor adverse effects and adjust dosing as needed to maximize patient safety. ➔ D risk level study (very high level of risk)
Number of participants included	36 patients maximum in total
Number of centres	6 centers (Hôpital Saint-Louis, Paris ; Hôpital Cochin, Paris ; Hôpital Avicenne, Bobigny ; Institut Gustave-Roussy, Villejuif; Institut Paoli-Calmettes, Marseille, CHU Nice – Hôpital Larchet, Nice)
Duration of the study	Enrolment period: 24 months Treatment period: up to 4 months Follow-up period: 12 months

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	Total study duration: 36 months.
Number of enrolments expected per site and per month	1 patient/center/4 months FPFV: July 2026 LPLV: April 2029
Statistical analysis	A one-stage Bayesian TITE-CRM design will be used for the trial primary objective. It allows sequential monitoring of data during the trial, using all current evidence for straightforward inference on safety and efficacy and decision making to select and evaluate the maximum tolerated dose. The TITE-CRM design will include up to 36 patients, with cohort size of 3, and dose level 1 as starting dose. Target maximum DLT rate for the definition of the optimal dose (OD) will be 33%. Safety rules will be implemented. The design skeleton will be calibrated to optimize the probability of correct selection of the MTD.
Funding sources	Main funding: PHRC-I Other funding: IHU THEMA
Data Safety Monitoring Board	A 3-member DSMB will review safety data.

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2 SCIENTIFIC JUSTIFICATION FOR THE STUDY

2.1 Hypothesis for the study

The study hypothesizes that the addition of Actinomycin D to Venetoclax is safe in patients with relapsed/refractory AML and will improve the response rate in this heavily pretreated population

2.2 Description of knowledge relating to the condition in question

Acute myeloid leukemia (AML) is a heterogeneous clonal myeloid neoplasm characterized by the uncontrolled proliferation of undifferentiated hematopoietic stem and myeloid progenitor cells, leading to ineffective hematopoiesis and bone marrow failure. Despite advances in the understanding of AML pathogenesis, this disease remains a hematologic malignancy with very poor prognosis, with approximately 18,000 new cases diagnosed annually in Europe and a five-year overall survival rate of only 17%[1].

The standard of care (SOC) for newly diagnosed AML involves an intensive chemotherapy regimen known as “7+3”, which combines cytarabine and an anthracycline (idarubicin or daunorubicin), followed by multiple cycles of high-dose cytarabine with or without allogeneic hematopoietic stem cell transplantation (HSCT)[19]. However, nearly one-third of AML patients fail to achieve remission following initial treatment, a condition referred to as primary refractory AML, which carries a dismal prognosis with a median survival of less than one year[3]. Even among patients who do achieve remission, relapse rates remain high, and most relapsed cases do not harbor actionable driver mutations (such as FLT3 or IDH1/2), further complicating treatment strategies[4].

For older or frail patients who are ineligible for intensive chemotherapy, treatment typically involves hypomethylating agents (HMAs) such as azacitidine or decitabine, often in combination with the BCL-2 inhibitor Venetoclax. This combination has improved remission rates and survival in frontline therapy[6]. However, patients who relapse or become refractory to HMA-based therapies face extremely poor outcomes, with median survival under 12 months[7]. Given the limited effectiveness of salvage therapies, novel treatment strategies that do not rely on traditional chemotherapy or HMA backbones are urgently needed to improve outcomes in R/R AML patients

2.3 Summary of relevant pre-clinical experiments and clinical trials

Actinomycin D and Venetoclax show strong preclinical and clinical synergy in AML.

Venetoclax, an orally available BCL-2 inhibitor, received accelerated FDA approval in 2018 for AML treatment. By mimicking BH3-only proteins, Venetoclax selectively binds and inhibits BCL-2, disrupting apoptotic resistance and leading to cytochrome C release and apoptosis[8]. However, resistance to Venetoclax is frequently associated with MCL-1 upregulation, which supports leukemic cell survival[11].

Actinomycin D, an RNA Polymerase I inhibitor, induces nucleolar stress and TP53 activation, leading to cellular senescence and apoptosis[9]. Additionally, Actinomycin D has been shown to downregulate MCL-1 expression, thereby overcoming Venetoclax resistance mechanisms and enhancing AML cell apoptosis. This effect is further supported by preclinical evidence demonstrating that Actinomycin D decreases MCL-1 levels while simultaneously increasing Noxa expression, a pro-apoptotic protein that enhances BH3-mimetic sensitivity[10].

Preclinical studies indicate that combining Actinomycin D with Venetoclax induces synergistic cytotoxicity, reinforcing its potential in AML treatment. Notably, recent studies have revealed that Actinomycin D targets mitochondria in AML cells, particularly in the context of NPM1-mutated

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AML, leading to the release of mitochondrial DNA and activation of cyclic GMP-AMP synthase signaling. This cascade enhances reactive oxygen species (ROS) production, which in turn re-establishes promyelocytic leukemia (PML) nuclear body formation—a key regulatory mechanism of mitochondrial fitness and senescence. This process ultimately reactivates TP53 signaling, inducing senescence and apoptosis in AML cells.

There is also preclinical evidence of Actinomycine D action in NPM1wt AML, by triggering nucleolar stress that reactivates the p53 pathway without requiring cytoplasmic export of NPM1. This makes leukemic cells, especially those with intact TP53 and high ribosome biogenesis, more prone to apoptosis[20].

Moreover, dual targeting of mitochondria with Actinomycin D and Venetoclax has demonstrated enhanced clearance of AML cells in vivo, leading to prolonged survival in preclinical models. These findings highlight a previously unrecognized role for mitochondrial dysfunction in AML pathogenesis and suggest that Actinomycin D enhances Venetoclax efficacy by enforcing a mitochondrial/ROS/PML/TP53 senescence axis. The combination has shown superior therapeutic efficacy compared to either agent alone, with a profound impact on leukemic cell viability and survival outcomes[12].

Clinical experience with the combination of actinomycin D and venetoclax in AML remains limited up to 70 reported patients in relapsed/refractory or post-transplant settings, where safety was manageable and response rates around 30–50% were observed, supporting further investigation. Retrospective analyses have shown that Actinomycin D as a single agent can lead to CR/CRi rates of up to 44% in patients with R/R AML[9,13]. Additionally, retrospective studies of Actinomycin D combined with Venetoclax in R/R AML patients have demonstrated ORR rates ranging from 26% to 75%, even in Venetoclax-exposed patients[14,15]. Importantly, these studies reported no treatment-related mortality, highlighting the tolerability and safety of this combination.

Collectively, these findings support the clinical evaluation of Actinomycin D and Venetoclax in R/R AML, particularly in patients who have exhausted conventional treatment options. This phase 1 study aims to assess the safety, feasibility, and preliminary efficacy of this regimen while exploring potential biomarkers of response, to guide future therapeutic strategies in this high-risk population.

2.4 Justification for the Selected Dosages and Schedules of Actinomycin D and Venetoclax

Actinomycin D :

Actinomycin D is an FDA-approved antitumor antibiotic used for several pediatric and adult solid tumors (e.g., Wilms tumor, rhabdomyosarcoma, Ewing sarcoma). However, it is not approved for use in AML. According to the product SMPC (e.g., Vidal), the recommended dosage varies between 10 and 15 µg/kg/day depending on the indication. In relapsed/refractory AML (R/R AML), retrospective studies such as those by Zucenka et al. (2020, 2021) report clinical activity at 12.5 µg/kg/day for 3 days, with manageable toxicity. Based on this clinical precedent and the absence of prospective safety data in combination with Venetoclax, the ACTIVATE protocol proposes an inter-patient dose-escalation scheme of 1 to 3 days of Actinomycin D (at 12.5 µg/kg/day), starting on Day 3 of the cycle, initiating the evaluation with the lowest level: 1 day. This design allows for stepwise toxicity evaluation and determination of the optimal duration of exposure within this combination strategy.

Venetoclax:

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Venetoclax is approved in AML in combination with azacitidine or decitabine, with dose adjustments required in case of co-administration with CYP3A4 inhibitors such as azole antifungals. Pharmacokinetic simulations from Bhatnagar et al. (Cancer Chemother Pharmacol, 2021; PMID: 33398386) demonstrated that co-administration of posaconazole with 50–100 mg/day of Venetoclax yields exposures equivalent to the standard 400 mg dose without CYP3A4 inhibition. In addition, multiple clinical studies using Venetoclax in combination with intensive chemotherapy (e.g., FLAG-IDA, IDA-AraC) have safely used a daily dose of 70 mg under azole prophylaxis with promising efficacy and tolerability (Willekens et al., 2024 [PMID: 40000842]; Kuusanmäki et al., 2020; Ghilardi et al., 2025).

The 7-day duration of Venetoclax was selected based on recent data showing comparable response rates between short (≤7 days) and extended (14–28 days) schedules, with reduced myelosuppression and infection risk when combined with chemotherapy or cytotoxic agents (Willekens et al., Haematologica 2024). This shorter duration is also consistent with emerging international practices in high-risk AML subgroups.

Cycle duration of 28 days was chosen to match standard hematologic treatment cycles, enabling consistent monitoring of hematologic recovery, response assessment, and toxicity across centers. Furthermore, following standard practice, cycle windows may be extended beyond D28 upon prolonged myelosuppression, for monitoring until recovery; treatment administration will nevertheless be capped at 28 days per cycle. It also facilitates comparisons with existing AML protocols.

2.5 Description of the population to be studied and justification for the choice of participants

- *Criteria related to the disease*
    - Adult patients (≥18 years) diagnosed with relapsed or refractory acute myeloid leukemia (R/R AML) who have failed at least one prior line of therapy. Patients from any oncogenetic subgroup will be eligible for inclusion, as the pharmacodynamics of Actinomycin D and Venetoclax are not associated with a specific AML molecular subtype. Given the poor prognosis associated with R/R AML, particularly in patients who have relapsed following intensive chemotherapy or hypomethylating agents plus Venetoclax, this population represents a critical unmet medical need. The inclusion of patients with prior exposure to Venetoclax allows for evaluation of the potential Venetoclax resistance-overcoming properties of Actinomycin D.
    - Exclusion criteria will include patients with acute promyelocytic leukemia (APL) and active CNS involvement.
    - Myeloid sarcomas are excluded because evaluation of response requires different methods compared to AML.

- *Organ failures and functional insufficiency*
    - Patients must meet eligibility criteria, including adequate organ function and ECOG performance status ≤2, to ensure they can tolerate study treatment.

2.6 Identification and description of the investigational medication or medications

The investigational medication is Actinomycin D and its association with Venetoclax. Further details are provided in Section 7 below.

2.7 Description and justification of the dosage, route of administration, administration schedule and treatment duration

Dose Levels for Actinomycin D are based on protocol available with safety data[15] in keeping with FDA guidance for the co-development of two drugs[18]. Actinomycin D will start on d3, after initial ramp up dosage of venetoclax, up to 3 days.

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Venetoclax will be given with a ramp up dosage over 3 days (20-50-70mg) until the daily dose of 70mg/d in patient with azole antifungal prophylaxis starting at D1. Venetoclax is given until D7 in each cycle to avoid further hematological toxicity[21].

This treatment is administered in hospitalized patients.

Further information about the administration of the product is given in **section 7**.

2.8 Summary of the known and foreseeable benefits and risks/burdens for the research participants

Foreseeable benefits

The ACTIVATE trial aims to provide new therapeutic options for patients with relapsed/refractory AML, a population with limited treatment alternatives and poor prognosis. Participants in the study may benefit from potentially improved response rates. The combination of Actinomycin D and Venetoclax has shown synergistic anti-leukemic effects, which could lead to higher remission rates and prolonged survival. Furthermore, identifying biomarkers of response may contribute to personalized medicine approaches, optimizing treatment strategies for future patients. Patients who achieve a response with the experimental treatment will be able to undergo an allogenic stem cell transplantation with the hope of curing their leukemia.

Foreseeable risks and burdens

The cautious dose-escalation design described in the **section 2.3.2** and the Bayesian approach, with a continuous reassessment of safety endpoints, enable to limit the risks related to the introduction of this drug.

Participation in the ACTIVATE trial introduces potential risks and burdens associated with the investigational treatment:

- Actinomycin D and Venetoclax can cause hematologic toxicities, including prolonged myelosuppression, neutropenia, and thrombocytopenia, which may increase the risk of infections and bleeding complications, in patients with already impaired bone marrow function by the active leukemia.

- Non-hematologic toxicities of Actinomycin D such as gastrointestinal disturbances (nausea, diarrhea, mucositis) are also expected. To mitigate this risk, the treatment will be performed in hospitalization with close monitoring of side effect.

- Patients will require frequent hospital visits and hospitalisation, laboratory assessments, bone marrow aspiration increasing the overall burden.

- The potential for tumor lysis syndrome (TLS) necessitates strict hydration protocols and prophylactic measures.

Despite these risks, the study is designed to closely monitor adverse effects and adjust dosing as needed to maximize patient safety.

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3 OBJECTIVES

3.1 Primary objective

To identify the maximum tolerated dose (MTD) of Actinomycin D / Venetoclax combination in patients with R/R AML.

3.2 Secondary objectives

To characterize the safety of the combination of Actinomycin D and Venetoclax in patients with R/R AML.

To preliminarily assess the anti-leukemia efficacy of the combination of Actinomycin D/ Venetoclax treatment.

To assess the combined pharmacokinetics and pharmacodynamics of Actinomycin D/ Venetoclax treatment in leukemia cells of peripheral blood and bone marrow during both stages of the clinical trial.

3.3 Exploratory objectives

Outcomes of patients enrolled in ACTIVATE will be compared with those of the ETHEMA (NCT: 05326919) cohort, using an appropriate matching strategy to account for baseline characteristics.

To assess the combined pharmacokinetics of Actinomycin D/ Venetoclax treatment.

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4 STUDY DESIGN

4.1 Study endpoints

4.1.1 Primary endpoint

Dose Limiting Toxicity (DLT)

DLT will be defined during a safety observation period corresponding to the first cycle of Actinomycin D/Venetoclax treatment, as:

- Death of any cause (with the exception of deaths caused by clear evidence of disease progression, defined as a leukocytosis greater than 50 × 10⁹/L with no evidence of toxicity)
- Prolonged myelosuppression on day 42 from start of therapy or later without evidence of leukemia defined by the absence of platelet recovery above 50 × 10⁹/L without transfusion support within the last week and the absence of neutrophil recovery with ANC > 1 × 10⁹/L
- Grade ≥ 3 non- hematological toxicity with the exception of:
  - Grade 3 or 4 infections
  - Grade 3 or 4 fever with neutropenia
  - Grade 3 nausea, vomiting or diarrhea that can be managed to ≤ Grade 2 within 72 hours of symptomatic treatment
  - Grade 3 mucositis or stomatitis that can be managed to ≤ Grade 2 within 72 hours of symptomatic treatment
  - Grade 3 tumor lysis syndrome that resolves within 72 hours of onset with medical treatment
  - Grade 3 or 4 hemorrhage

The first cycle of treatment usually corresponds to 28 days in duration, but may be prolonged up to 42 days, in the case of prolonged treatment-related cytopenia. As described above, the absence of recovery from aplasia at day 42 will qualify as a DLT, thus completing the primary observation. Conversely, in the absence of aplasia at day 28, and in the absence of other DLT event, the patient will be considered as DLT-free (complete observation) and treatment will be carried on to the next cycle without waiting further, for ethical reasons.

Probability of DLT should not exceed 33% at the end of the first cycle of Actinomycin D/Venetoclax.

4.1.2 Secondary endpoints

1) Assessment of safety

Adverse events (AE), Treatment emergent adverse events (TEAE), treatment related TEAE, TEAES leading to treatment discontinuation, serious adverse events (SAE): frequency, duration and severity of each eventand deaths should be documented during phase 1 and 2 of the trial.

Safety outcome measures will be assessed continuously during the whole study participation. Monitoring of ECGs and clinical laboratory values are integral to safety assessment.

AE, TEAE and Treatment-related TEAE will be evaluated according to the NCI CTCAE version 6.0. Serious adverse events will be defined per protocol according to EMA/ICH criteria. Causality relationship with the experimental treatment should be assessed by the investigator.

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Patients should be followed until all treatment related adverse event have returned to grade ≤ 1 or baseline or are deemed irreversible by the investigator.

2) Preliminary assessment of anti-leukemia efficacy

- **Complete response** rate including CR, CRi (CR with incomplete hematologic recovery, meaning CR with platelet count <100,000/µL or absolute neutrophil count <1000/µL) and CRh (CR with partial hematologic recovery, meaning CR not fulfilling CR or CRi peripheral blood count criteria but with platelet count >50,000/µL AND absolute neutrophil count >500/µL).
- **Morphologic leukemia-free state** (MLFS): blastic criteria of response on the bone marrow aspirate, absence of extramedullary disease and no requirement of hematologic recovery.
- **Partial response** (PR): absence of circulating blasts; absence of extramedullary disease; decrease of bone marrow blast percentage between 5 and 25% and decrease of pretreatment bone marrow blast percentage by at least 50%.
- **Overall response rate** (ORR) being defined as CR, CRi and CRh, MLFS and PR rates.
- **Event-free survival** (EFS); here events are defined as:
    - Non achievement of hematologic response including CR, CRi, CRh
    - Hematologic relapse or progressive disease
    - Initiation of any subsequent anti-leukemic therapy (excluding hydroxyurea)
    - Deaths prior to one of the events mentioned above

Reasons for subsequent therapy initiation (including intolerance, resistance, hematologic relapse, unsatisfactory MRD evolution), its type (intensive, less-intensive, best supportive care) and its frame (clinical trial, off-label or compassionate use) will be prospectively recorded. Bone marrow aspirate will be performed at the end of each cycle.

- **Progressive disease** (PD):
    - >50% increase in marrow blasts over baseline (a minimum 15%-point increase is required in cases with <30% blasts at baseline); or a persistent marrow blast percentage of >70% over at least 3 months, (without at least a 100% improvement in ANC to an absolute level >0.5x10^9/L and/or platelet count to > 50 x 10^9/L, non-transfused);
    - Or a new extramedullary disease

- **Duration of response** (DOR),
- **Relapse free survival** (RFS)
- **Overall survival** (OS)
- **Early mortality rates after 1 cycle of treatment** (day 28-42) () and **after 2 cycles** (day 56-84) ()
- Incidence of **subsequent allogeneic HSCT**, overall and in responding patients specifically
- **Molecular Residual Disease** (MRD) will be evaluated as follows: Peripheral blood RQ-PCR for NPM1 patients[16], BM RQ-PCR for t(8;21) and inv(16) patients[17], otherwise FCM.

Apart from specific mentions, all these hematological endpoints will be defined according to the last recommendation of the European Leukemia Network[19].

3) Pharmacokinetics and Pharmacodynamics

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To assess the combined pharmacokinetics and pharmacodynamics of Actinomycin D/Venetoclax treatment in leukemia cells, with research of biomarkers of response. They will be done on peripheral blood, because repeated bone marrow aspiration over a short-time period is unethical at D3 and D7.
For pharmacokinetic analysis, the dose of venetoclax will be administered orally.

Blood samples will be obtained:
- on day 3 pre-dose
- on day 7 pre-dose

Blood samples will be collected into heparinized tubes. Immediately after collection, tubes will be inverted several times and kept at approximately 4 °C until centrifugation. The tubes will be centrifuged for 10 min at 1500×g at 4°C within 60 min of collection to separate plasma. The plasma will be separated into 2 aliquots in polypropylene screw-cap tubes and placed at -80°C until analysis.
All samples will be labeled with patient identification, trial identification, sample number, and actual date and time at which the sample was collected.

4.1.3 Secondary endpoints

Outcomes of patients enrolled in ACTIVATE will be compared with those of the ETHEMA (NCT: 05326919) cohort, using an appropriate matching strategy to account for baseline characteristics.

Combined pharmakinetics of Actinomycin D/Venetoclax treatment will be done on peripheral blood at D3 and D7.

4.2 Description of research methodology

4.2.1 Design of the study

Dose-escalation phase 1 open label, multicenter clinical trial following a one-stage adaptive Bayesian time-to-event continual reassessment method (TITE-CRM) design, to identify the maximum tolerated dose of Actinomycine D in combination with Venetoclax days 1-7.

The design will be applied to a single, specific population: adult patients with a confirmed diagnosis of relapsed or refractory AML, regardless of prior treatments or number of relapses.

Number of subjects: 36 patients in total.

4.2.2 Number of participating sites

This French multicenter study will involve a total of 6 centers. Patients will be recruited in hospitals by hematologists expert in AML management. All patients will be hospitalized before the initiation of the experimental therapy.

4.2.3 Identification of participants

The participants in this research will be identified as follows:
Site number. (3 digits) - Sequential enrolment number for the site (4 digits).
This reference number is unique and will be used for the entire duration of the study.

4.2.4 Randomisation

The design does not include a randomization procedure.

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Dose assignments will be performed centrally by the statistical team at the Service de Biostatistique et Informatique Médicale at Hôpital Saint Louis, Paris.

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5 IMPLEMENTATION OF THE STUDY

Whose consent must be obtained	Who informs the individuals and collects their consent	At what point the individuals are informed	At what point the consent is obtained
The individual participating in the study	The principal investigator or collaborating physician declared and trained in the study (Haematologist)	72 hours before sreening visit	After a reflection period of 72 hours at the latest at the time a specific procedure for the study is implemented

The visits will be conducted by an investigator expert in acute myeloid leukemia management.

Patients will most often be hospitalized at screening visit and will always be hospitalized at the beginning of the induction course.

5.1 Screening visit

The screening visit takes place between 1 day and no later than 14 days before the baseline visit and is intended to collect the patient informed consent and to assess eligibility criteria.

Specifically, the visit will record:
- Signed informed consent form (ICF)
- Medical and disease history and demographics
- Physical examination
- Height and weight
- Vital signs determination
- ECOG performance status evaluation
- HCT-CI
- Assessment of prior and concomitant medications, including medications received within 14 days prior to screening.
- 12-lead ECG
- Echocardiography for LVEF
- Clinical laboratory tests: complete blood count, peripheral blood smear, glucose, sodium, potassium, calcium, phosphorus, magnesium, uric acid, blood urea nitrogen (BUN), creatinine level, troponin, AST, ALT, alkaline phosphatases, γGT, bilirubin, LDH, lipase, fibrinogen, partial thromboplastin time and prothrombin time [PT]/international normalized ratio [INR]
- Creatinine clearance according to MDRD
- Viral Serology if not performed within the last 3 months: HIV, HVC, HVB, CMV, EBV, HTLV1
- Serum β-hCG test
- Hematological specialized assessment:
    - Bone marrow aspiration and peripheral blood for confirmation of diagnosis and baseline characteristics of the AML. A bone marrow biopsy will be performed only if aspiration inadequate. Bone marrow cytogenetic and routine molecular biology can be performed in local laboratories.

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- Bone marrow aspiration and peripheral blood for determination of LAIP (baseline for flow MRD) and *NPM1* transcript quantification (baseline for *NPM1* MRD in *NPM1*-mutated patients)

- Biomarkers:
  - Targeted sequencing of a panel of genes recurrently mutated in AML

Enrollment Procedure

Upon approval by the Coordinating Investigator and signature of the informed consent form, the patient will be enrolled by the Investigator through the eCRF.

The investigational team will inform in advance the statistical team of each scheduled enrolment; an on-call statistician will be identified to perform the upcoming dose assignment computation. Moreover, on the day of inclusion, the statistical team will be informed by an automated mailing from the eCRF tool of the effective inclusion of the new patient. Upon receiving this inclusion email, the on-call statistician will compute the dose level assigned to a new patient using the dose-finding TITE-CRM algorithm based on previous patients' updated DLT data extracted from the eCRF (see statistical details in **Section 12**).

Thus, **the DL of the study drug assigned to the patient being enrolled will be provided to the investigator via email** upon completing the enrolment procedure on the eCRF.

Details and outputs of the dose-assignment computations will be recorded for each included patient for traceability.

5.2 Follow-up visits

5.2.1 During cycle of treatment

Visits during one cycle of treatment will take place at D1, D2, D3, D7, D15 and D22 of the cycle.

For each cycle, patients will receive:

- Venetoclax daily days 1-7,
- Actinomycin D by short IV infusion 12.5 µg/kg/d for up to 3 days starting on D3.

Venetoclax will be given with a ramp up dosage over 3 days (20-50-70mg) until the daily dose of 70mg/d is achieved in patient with azole antifungal prophylaxis starting at D1. Venetoclax is given until D7 in each cycle to avoid further hematological toxicity.

Three dose schedules of actinomycin D will be explored in induction phase, starting at DL1:

- DL1: Venetoclax daily days 1-7 and Actinomycin D 12.5 µg/kg/d at day 3,
- DL2: Venetoclax daily days 1-7 and Actinomycin D 12.5 µg/kg/d at days 3-4,
- DL3: Venetoclax daily days 1-7 and Actinomycin D 12.5 µg/kg/d at days 3-5

Dose escalation will be conducted according to a dose finding design with sequential dose assignment with cohorts of size 3, the number of patients eventually assessed at each dose level depending on the DLT.

Visits will also report:

- Physical examination
- Vital signs determination
- ECOG performance status evaluation (at D1)

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- Assessment of prior and concomitant medications
- Clinical laboratory results, at each visit: complete blood count, peripheral blood smear, glucose, sodium, potassium, calcium, phosphorus, magnesium, uric acid, blood urea nitrogen (BUN), creatinine level, AST, ALT, alkaline phosphatases, γGT, bilirubin, LDH, lipase, fibrinogen, partial thromboplastin time and prothrombin time (PT)/international normalized ratio (INR)
- AE/SAE evaluation
    - PD/PK: at D3 and D7

5.2.2 End of cycle visit (D28 to D42 of cycle)

Visits for each end of cycles will take place between since D28 as soon as hematological recovery is obtained and no longer than D42 and the following procedures will be performed / variables will be collected:

- Physical examination
- Vital signs determination
- ECOG performance status evaluation
- Assessment of prior and concomitant medications
- Clinical laboratory results: complete blood count, peripheral blood smear, glucose, sodium, potassium, calcium, phosphorus, magnesium, uric acid, blood urea nitrogen (BUN), creatinine level, troponin, AST, ALT, alkaline phosphatases, γGT, bilirubin, LDH, lipase, fibrinogen, partial thromboplastin time and prothrombin time (PT)/international normalized ratio (INR)
- Specialized hematological assessment:
    - Bone marrow aspiration for assessment of response according to ELN 2022 criteria. A bone marrow biopsy will be performed only if aspirate is inadequate
    - Bone marrow aspiration for flow based MRD (sent to the central hematology lab of Saint-Louis hospital)
    - Bone marrow aspiration and peripheral blood for *NPM1* transcript MRD in *NPM1* mutated patients
- Biomarkers: bone marrow aspiration for biomarkers (PD)

5.2.3 Post treatment visits

These visits will take place at the end of each consolidation cycle, before allogeneic stem cell transplantation (HSCT), 3 months after the last chemotherapy course (consolidation or HSCT conditioning regimen) and every 3 months until the end of study visit, where the following procedures will be performed / variables will be collected:

- Physical examination
- Vital signs determination
- Assessment of prior and concomitant medications:
    - Consolidation chemotherapy will be documented
    - In transplanted patients, the origin of the donor, stem cell source and intensity of conditioning regimen will be documented
- Clinical laboratory results: complete blood count, peripheral blood smear, glucose, sodium, potassium, calcium, phosphorus, magnesium, uric acid, blood urea nitrogen (BUN),

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creatinine level, AST, ALT, alkaline phosphatases, γGT, bilirubin, LDH, lipase, fibrinogen, partial thromboplastin time and prothrombin time (PT)/international normalized ratio (INR)
- If performed, results from bone marrow aspiration/biopsy will be recorded
- If performed, results from local bone marrow or peripheral blood MRD will be recorded

After completion of the two cycles, further therapeutic decisions will be made according to standard clinical practice at the discretion of the treating physician. These subsequent treatments are not part of the investigational study treatment. In responding patients, allogeneic hematopoietic stem cell transplantation may be considered if clinically appropriate.

5.2.4 Relapse

If the patient relapses, an additional visit will take place where the following procedures will be performed / variables will be collected:

- Physical examination
- Vital signs determination
- ECOG performance status
- Assessment of prior and concomitant medications
- Clinical laboratory results: complete blood count, peripheral blood smear, glucose, sodium, potassium, calcium, phosphorus, magnesium, uric acid, blood urea nitrogen (BUN), creatinine level, AST, ALT, alkaline phosphatases, γGT, bilirubin, LDH, lipase, fibrinogen, partial thromboplastin time and prothrombin time (PT)/international normalized ratio (INR)
- Hematological specialized assessment with its date, enabling to specify the date of the relapse:
    - Bone marrow aspiration and peripheral blood. A bone marrow biopsy will be performed only if aspiration inadequate. Bone marrow cytogenetic and routine molecular biology can be performed in local laboratories
    - Pathology report in case of extramedullary relapse

5.3 Last study visit

During this visit, the following procedures will be performed / variables will be collected:

- Physical examination
- Vital signs determinations
- ECOG performance status evaluation
- Assessment of prior and concomitant medications
- Clinical laboratory results: complete blood count, peripheral blood smear, glucose, sodium, potassium, calcium, phosphorus, magnesium, uric acid, blood urea nitrogen (BUN), creatinine level, AST, ALT, alkaline phosphatases, γGT, bilirubin, LDH, lipase, fibrinogen, partial thromboplastin time and prothrombin time (PT)/international normalized ratio (INR)
- Results from any bone marrow aspiration/biopsy performed since the last post induction visit will be recorded
- Results from any local bone marrow or peripheral blood MRD assessment performed since the last post induction visit will be recorded

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5.4 Early termination visit

Patients are allowed to withdraw their participation in the research at any time and for any reason. The investigator can temporarily or permanently withdraw a participant from the study for any safety reason or if it is in the participant's best interests.

If a participant wishes to withdraw his consent from the study, we will use the following strategies to minimize the impact on the validity of the trial meanwhile respecting the participant's right to withdraw. We will seek a better understanding of the patient's wishes and offer the following alternatives to complete withdrawal:

1) Discontinue in-person follow-up but allow telephone follow-up;
2) Discontinue in person and by telephone follow-up but allow access to medical records and to use personal data to be able to get eventual death status from the website https://www.deces-en-france.fr/

In any case, all the data that would have been collected before the patient's consent withdrawal will be kept in. The early termination visit will be similar to the last study visit (see **section 5.4**).

5.5 Expected length of participation and description of the chronology and duration of the study.

Maximum period between screening and enrolment: 14 days
Duration of enrolment period: 24 months
Duration of treatment : 2 cycles of 42 days at most
Duration of participation for each participant: 12 months
Treatment period: up to 4 months
Total study duration*: 36 months *End of trial is defined as the last visit of the last patient.

5.6 Table or diagram summarising the chronology of the study (R = research / C = care)

Study Visit	Screening visit	Cycle 1 & 2								D28-D42 of each cycle	Quarterly visit until end of study (Month 12)
		D1	D2	D3	D4	D5	D7	D15	D22		
Inclusion and exclusion crit.	R										
Informed Consent	R										
Actinomycin treatment				R	(R)[8]	(R)[8]					
Venetoclax treatment		R	R	R	R	R	R				
Medical history & demographics	R										
Physical exam	R	C	C	C			C	C	C	R	C
Height and weight	R	C								R	C
Vital signs	R	C	C	C			C	C	C	R	C
ECOG PS	R	C								R	C
BM aspirate[1]	R/C									R	
Cytogenetics[1]	C									R	
Genetics[1]	C									R	
PD[2]/PK				R			R			R[7]	
Response Assessment[3]										C	
Hematology	C	C	C	C			C	C	C	R	C
Chemistry[4]	C	C	C	C			C	C	C	R	C
Pregnancy test[5]	R										
Viral serology[6]	R										
Creatinine clearance	C	C	C	C			C	C	C	R	
ECG	C	C								R	

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Echocardiography	R										
HCT-CI	R										
Prior and concomitant med.	R	C	C	C			C	C	C	R	C
Experimental treatment		R	R	R			R				
AE/SAE		R	R	R			R	R	R	R	R
Relapse											C
Alive status											C

[1] Additional BM aspirates with cytomorphology for response assessment (including MRD) are also mandatory:
- Before an allogeneic HCT if any.
- At any time, a suspicion of progression with a > 50% increase in peripheral blasts to >25 x 10^9 /L, or suspicion of extra-medullary disease.

[2]PD on peripheral blood, platelet-rich plasma, and BM when visits include BM aspirates. PK on PB.
[3]Peripheral blood RQ-PCR for NPM1 patients[16], BM RQ-PCR for t(8;21) and inv(16) patients[17], otherwise FCM.
[4] Including Ionogram, Ca, P, Mg Renal, Liver tests, lipase, glucose, lipid tests.
[5] restricted to women of childbearing potential.
[6] if not performed within the last 3 months: HIV, HVC, HVB, CMV, EBV and HTLV1.
[7] only PD analyses
[8]According to patient dose cohort

5.7 Distinction between standard care and study

This is an interventional study in the setting of relapse refractory AML. In this condition, there is no recommended treatment. Patients with relapse refractory AML are usually treated with best supportive care or enrolled in early phase clinical trial.

Interventions added by the studies are the following procedures:

- Actinomycin D administration in combination with Venetoclax administration,
- PD exams from peripheral blood samples (plasma, serum or cells)

TABLE: "Standard care" vs. "additional interventions" required specifically for the study

Interventions carried out for the research purposes	**Interventions, procedures and treatments associated with <u>standard care</u>**	**Interventions, procedures added for <u>research purposes</u>**
Visits	Standard care	Screening
Treatments		Administration of Actinomycin D in association with Venetoclax.
Procedures	Central IV line	Pregnancy test for WOCBC Echocardiography
Blood samples	Monitoring of hematology and chemistry tests. PB MRD for *NPM1* transcripts in *NPM1* mutated patients	Viral serologies (5mL) PD and PK monitoring at D3 (10 mL), D7 (10 mL) and end of cycle (only PD). RQ-PCR for NPM1 patients (centralized) at end of cycle Up to 2 cycles.
Bone Marrow	Aspiration at screening (diagnosis) for cytology smears, cytogenetics (1 mL), flow cytometry (1 mL)	Additional volume (2 mL EDTA) at screening and end of each cycle evaluation aspirations for biomarkers (centralized PD).

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	and molecular genetics (1 mL). Aspiration at end of cycle evaluation for cytology smears, flow cytometry (1 mL) and molecular genetics (1 mL).	RQ-PCR for t(8;21) and inv(16) patients, otherwise FCM (centralized) at end of cycle Up to 2 cycles.

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5.8 Biological samples collection

Cells from bone marrow and peripheral blood samples taken at protocol specified visits will be stored in a biological sample collection.

During the study the sample collection(s) will be stored at CRB Saint Louis (AP-HP) under the supervision of Pr Emmanuelle Clappier for no more than 20 years. The collection is intended to perform the biomarker and PD analyses of the study.

At the end of the study, remaining samples will be kept and may be used for further analyses not described in the initial protocol, but which may be useful for investigation of acute myeloid leukemia in light of advances in scientific knowledge, provided the participant is informed and does not oppose this, as stated in the informed consent form.

If the samples are kept at the end of the study, the sample collection will be declared to the ministry of research and to the director of the competent regional healthcare authority - (Article L. 1243-3 of the *Code de la Santé Publique* [French Public Health Code]).

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6 ELIGIBILITY CRITERIA

6.1 Inclusion criteria

- Patients aged 18 years old or more
- Patients with confirmed diagnosis of R/R AML according to the 2022 ELN recommendations who have failed available standard therapies, are not eligible for standard therapeutic options, or for whom no appropriate standard treatment option is considered available according to the investigator's judgment
- WBC < 20,000/mm3 (previous HY is allowed to control WBC count)
- ECOG performance status ≤2
- Any prior treatment-related toxicities resolved to ≤ Grade 1 prior to first study dose, with the exception of ≤ Grade 2 neuropathy or alopecia
- AST and ALT ≤3.0 x upper the limit of normal (ULN) and total and direct serum bilirubin ≤ 1.5 x ULN unless considered due to leukemia, in which case thresholds may be increased to ≤ 6.0 × ULN for transaminases and ≤ 3.0 × ULN for bilirubin.
- Estimated glomerular filtration rate (GFR) ≥ 50 mL/min/1,73m^2 according to the MDRD equation
- Posaconazole prophylaxis for invasive pulmonary aspergillosis (IPA)
- Written informed consent obtained prior to any screening procedures
- Patients affiliated or beneficiary of the National Health Insurance in France (AME excluded)

6.2 Exclusion criteria

- Acute promyelocytic leukemia
- Proven central nervous system leukemic involvement
- Prior allogeneic HSCT within the last 12 weeks or donor lymphocyte conditioning (DLI) within the last 4 weeks
- Graft-Versus-Host Disease (GVHD): Signs or symptoms of acute GVHD ≥ Grade 2, moderate or severe limited chronic GVHD or extensive chronic GVHD of any severity within the last 4 weeks prior to the first study dose. Patients must have been off all systemic immunosuppressive therapy for at least 2 weeks prior to enrolment. Patients may receive as GVHD treatment corticosteroid doses ≤ 10 mg of prednisone daily.
- Concurrent therapy with any cytotoxic drug within 3 weeks before the first study dose. Only hydroxyurea for the control of blood counts is permitted.
- Adults weighting less than 40 kg
- Uncontrolled systemic fungal, bacterial, or viral infection (defined as ongoing signs/symptoms related to the infection without improvement despite appropriate treatment)
- HBV or HCV hepatitis with a detectable viral load or any ongoing anti-viral treatment
- HIV with a detectable viral load or any ongoing anti-viral treatment
- Other uncontrolled or active malignant disease within prior 12 months (excluding myelodysplastic syndrome; myeloproliferative neoplasm, cutaneous basal cell carcinoma, "in-situ" carcinoma of the cervix or breast, or other local malignancy excised)
- Inability to swallow
- Known malabsorption syndrome or other condition that may significantly impair absorption of oral study medications
- Administration of any therapy that is considered to be investigational (i.e., used for non-approved indications(s) or in the context of a research investigation; anti-leukemic therapy that is considered investigational, including agents used off-label or under compassionate use programs) within 3 weeks or 5 drug half-lives (whichever is longer) prior to the first dose of study drug
- Any concurrent severe and/or uncontrolled medical condition, which could compromise participation in the study

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- Adult subjects to a legal protection order or unable to give their consent (*tutelle, curatelle, sauvegarde de justice, habilitation familiale, mandate de protection future*)
- Persons deprived of their freedom by judicial or administrative decision, person hospitalized without their consent by virtues of articles L 3212-1 and L3213-1 and who are not subject to the provisions of article L 1121-8
- Females who are pregnant or breastfeeding
- For males whose sexual partner is a woman of childbearing potential: unwillingness or inability of the man or woman to use a highly effective contraceptive method for the entire treatment period and for at least 6 months after completion of protocol treatment
- For heterosexually active woman of childbearing potential: unwillingness or inability to use a highly effective contraceptive method for the entire treatment period and for at least 6 months after the administration of the final dose of study medication
- Known allergies to investigational medicinal products or to posaconazole
- Concomitant therapy with potent inhibitors or substrates (listed in addenda 18.7) of CYP3A
- Concomitant medication with St. John's wort preparations
- Concomitant medication with HMG-CoA reductase inhibitors (simvastatin, lovastatin, and atorvastatin)

Highly effective contraception methods include: combined (estrogen and progestogen containing) or progestogen-only hormonal methods associated with inhibition of ovulation, intra-uterine device, surgical sterilization (including bilateral tubal occlusion, partner's vasectomy), or sexual abstinence if this is the preferred and usual lifestyle of the patient. In accordance with the venetoclax SmPC, women using hormonal contraception must also use an additional barrier contraceptive method during treatment and for the applicable post-treatment contraception period.
Women are not regarded as of childbearing potential if they are post-menopausal (at least 2 years without menses) or are surgically sterile (at least 1 month before enrolment).

Female patients must not donate or retrieve, for their own use, ova from the time of screening and throughout the treatment period, and for 12 weeks after the administration of the final dose of study medication.

Female patients must agree not to breastfeed from the time of screening and throughout the protocol period, and for (7 half-lives) days after the administration of the final dose of study medication.

Male patients must not freeze or donate sperm starting at screening and throughout the treatment period and 3 months after the administration of the final dose of study medication.

6.3 Recruitment procedure

Patients will be recruited at hospital in onco-hematology department by haematologists also investigator of the study. A patient information form will be provided to explain the study and an appropriate time will be given to the patient before signing the consent form.

Total number of participants to be included	36 patients
Number of centers	6 centers
Enrolment period (months)	24 months
Number of participants/center	4 to 8
Number of participants/center/month	0,25 center/ month

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6.4 Termination rules

6.4.1 Criteria and procedures for prematurely terminating the study treatment

Several situations are possible:

- Temporary suspension of treatment: the investigator must document the reason for suspending and resuming the treatment in the participant's source file and the case report form (CRF).
- Premature discontinuation of treatment, but the participant remains enrolled in the study until the end of their participation.
- Premature discontinuation of treatment and withdrawal from the study.

The investigator must:

- Document the reason(s),
- Collect the assessment criteria at the time of ending participation in the study, if the participant agrees,
- Schedule a follow-up for the participant, particularly in case of a serious adverse event.

If, during the course of his/her participation in the study, the participant presents one non hematological grade ≥ 3 AE that is attributable to the study product in the investigator's opinion, then the study product must be interrupted until toxicity resolves or improve to grade 1, and the participant will continue to be monitored for the study.

In the case of severe adverse events, the investigator must notify the sponsor and follow up the participant until the adverse resolution if it occurred before.

Notification of a serious adverse event must be sent by email (eig-vigilance.drc@aphp.fr) to the sponsor. The Data Safety Monitoring Board can specify and/or validate the follow-up methods.

6.4.2 Population-level safety stopping rules

In addition to the individual treatment discontinuation rules described above, population-level safety monitoring will be performed throughout the study in order to detect potential unexpected toxicity signals.

Enrollment and/or dose escalation may be temporarily suspended if safety concerns arise during the study.

Potential triggers for safety review include:

- occurrence of treatment-related death,
- occurrence of unexpected Grade 4 non-hematologic toxicity related to study treatment,
- occurrence of multiple serious adverse events suggesting an unfavorable safety profile,
- identification of a toxicity pattern inconsistent with the expected safety profile of the investigational treatment.

If such safety signals are identified, patient enrollment or treatment administration may be temporarily suspended while a safety review is conducted.

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Safety data will be reviewed by the Data Safety Monitoring Board (DSMB), which may recommend:

- continuation of the study without modification,
- modification of the dose escalation scheme,
- suspension of enrollment,
- or termination of the study.

These stopping rules are independent from the Bayesian TITE-CRM dose escalation algorithm, and are intended to ensure patient safety at the study population level.

6.4.3 Criteria and procedure for premature withdrawal of a participant from the study

Participants may exit the study at any time and for any reason. If a participant exits the study prematurely and withdraws consent, any data collected prior to the date of premature exit may still be used, upon their consent.

Participant lost to follow-up: the participant cannot be located. The investigator must make every effort to reconnect with the participant (and document his attempts in the source file), at least to determine whether the participant is alive or dead.

In case of serious adverse events, see the corresponding section on vigilance.

The case report form must list the various reasons why the participant has discontinued the study:

- ☐ Explicit withdrawal of consent
- ☐ Lost to follow-up.

6.4.4 Follow-up of participants following premature withdrawal from the study

If a participant discontinues the study, this will in no way affect their usual care for their condition. In the event of serious adverse events following premature discontinuation of treatment and participation of the patient in the study; see section **6.4.1**.

6.4.5 Procedures for replacing participants,

Participants who have been included but eventually do not receive any study treatment will be replaced.

Participants who have withdrawn their consent before the end of the evaluation window for reasons other than death or DLT will not be replaced. The dose finding design allows accounting for incomplete observations due to patients' discontinuation.

6.4.6 Full or partial discontinuation of the study

Early and temporary or definitive discontinuation of the study, early and temporary discontinuation of inclusions can be decided based on DSMB recommendations, and whether or not participants are allowed to complete their treatment or should immediately discontinue study drugs will all be based on DSMB recommendations. The DSMB will examine all safety data throughout the trial, including the design's pre-specified safety rules (such as the probability of DLT at dose level 1 (DL1) being greater than 95%).

AP-HP as sponsor or the Competent Authority (ANSM) or the Ethics committee (CPP) can prematurely discontinue all or part of the study, temporarily or permanently, further to the recommendations of the Data Safety Monitoring Board in the following situations:

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- first of all, if any suspected unexpected serious adverse reactions (SUSARs) are observed, requiring a reassessment of the benefit-risk ratio for the study.
- Similarly, AP-HP, as the sponsor, or the Competent Authority (ANSM) or the Ethics committee (CPP) may decide to prematurely discontinue the study due to unforeseen issues or new information about the product, in light of which the objectives of the study or clinical program are unlikely to be achieved.

AP-HP as sponsor reserves the right to permanently suspend enrolment at any time if it appears that the inclusion objectives are not met.

If the study is prematurely discontinued for safety reasons, the decision and justification will be provided by the sponsor (AP-HP) to the Competent Authority (ANSM) and to the CPP (Research Ethics Committee) within a period of 15 days, along with recommendations from the Data Safety Monitoring Board in the case of substantial modification.

In case of a protocol substantial modification is necessary, it should be shared to the Competent Authority and to the Ethics Committee in CTIS platform, without undue delay but not later than 15 days of the date of the temporary halt or early termination.

In case of occurrence of an unexpected event that could have a serious impact on the risk/benefit of the trial, this unexpected event should be shared without undue delay but not later than 15 days from the date the sponsor became aware of this event.

In case of implementation of urgent safety measures, they should be taken immediately and should be shared to the Competent Authority and to the Ethics Committee in CTIS platform, without undue delay but no later than 7 days of the date of the measure implementation.

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7 TREATMENT ADMINISTERED TO STUDY PARTICIPANTS

7.1 Description of the investigational medicinal products

Treatment will be administered in 2 consecutive 28-day cycles. Cycle windows may be extended beyond D28 upon prolonged treatment-related cytopenia, for monitoring until recovery.

Patients will receive two molecules:

- Venetoclax every day from days 1 to 7,
- and actinomycin D as a short IV infusion of 12.5 µg/kg/d for a maximum of 3 days from day 3.

Treatment initiation will occur during hospitalization in order to ensure close clinical and biological monitoring during the administration of study drugs and during the expected period of treatment-related cytopenia.

Participants will remain hospitalized at least until Day 15 of each treatment cycle, corresponding to the early period of treatment administration and monitoring for potential adverse events, including cytopenia-related complications, tumor lysis syndrome and infectious events.

Beyond Day 15, continuation of hospitalization or transition to outpatient management may be considered according to the investigator's clinical judgment, depending on the patient's clinical condition, hematological recovery and occurrence of treatment-related adverse events.

Additional hospitalization may be required at any time during the study if clinically indicated.

7.1.1 Investigational medicinal: Venetoclax

Venetoclax (VEN) belongs to the family of B-cell lymphoma-2 (BCL-2) inhibitors and has been approved and used in France as a treatment for chronic lymphocytic leukemia (CLL) and acute myeloid leukemia (AML).

Tablets are film-coated and contain 10mg or 50mgof VEN each. Venetoclax is available commercially under the name VENCLYXTO®.

Given that the elimination half-life of VEN is ~26 hours in human blood, we anticipate that administration of the drug once daily will ensure a sufficient blood concentration.

Venetoclax is administered orally with water in increasing doses over 3 days (20-50-70mg) for prevention of tumor lysis syndrome, up to a daily dose of 70mg/d in patients with azole antifungal prophylaxis starting on D1. Venetoclax is administered until D7 of each cycle to avoid further hematological toxicity[21,22].

At the start of Cycle 2, venetoclax administration will depend on the patient's disease status at the end of Cycle 1. In patients with persistent active disease, venetoclax will be administered following the same dosing schedule as in Cycle 1, including the 3-day ramp-up phase (20 mg–50 mg–70 mg) followed by 70 mg/day until Day 7. In patients achieving blast clearance, venetoclax may be initiated directly at the target dose of 70 mg/day without repeating the ramp-up phase.

Tumour Lysis Syndrome

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Venetoclax may induce tumor lysis syndrome as a consequence of the rapid drug-induced lysis of neoplastic cells. Blood uric acid levels, potassium, calcium, phosphate, and creatinine should be evaluated after initial treatment. Hydration with isotonic fluids and prophylaxis with allopurinol/rasburicase are recommanded to prevent hyperuricemia may minimize potential complications of tumour lysis syndrome.

7.1.2 Investigational medicinal: Actinomycin D

Actinomycin D belongs to the family of cytotoxic antibiotics and has been approved and used in France as a treatment for various cancers, including Wilms' tumor, rhabdomyosarcoma, Ewing's sarcoma, and gestational trophoblastic neoplasia.

The injectable formulation is available as a lyophilized powder for reconstitution, containing 500 mcg (0.5 mg) of ActD per vial. Actinomycin D is commercially available under the name COSMEGEN®. The latter can be used in this clinical study.

Given that the elimination half-life of ActD is ~36 hours in human blood, we anticipate that administration of the drug once will ensure a sufficient blood concentration.

Actinomycin D is administered from D3, after the initial increase in Venetoclax dose, for up to 3 days. Three actinomycin D regimens will be studied in the induction phase, defining 3 dose levels for the trial:

- DL1 (initial dose): Venetoclax daily from D1 to 7 and Actinomycin D 12.5 µg/kg/d at D3,
- DL2: Venetoclax daily from D1 to 7 and Actinomycin D 12.5 µg/kg/d on D3-4,
- DL3: Venetoclax daily from D1 to 7 and Actinomycin D 12.5 µg/kg/d from D3 to D5.

Dose Levels for Actinomycin D are based on protocol available with safety data[15] in keeping with FDA guidance for the co-development of two drugs[18].

Dose escalation will be conducted according to a dose finding design with sequential dose assignment with cohorts of size 3, the number of patients eventually assessed at each dose level depending on the DLT observations.

Dosing should be interrupted or doses should be reduced as follows:

Adverse events & reactions	Recommended action
Hematological AE grade 4	Continue cycle. Delay D1 of next cycle up to D42 in cycles 1-2. In responding patients beyond cycle 2 (cycle 3+), next cycle can be delayed up to D56 and started at the next lower DL
Non hematological grade ≥3 TEAE	Interrupt Actinomycin D and Venetoclax until toxicity resolves or improves to grade ≤ 2. Resume and at the next lower dose level of both drugs, except for the following TEAEs where treatment can be restarted at the same DL: oral mucositis. Next cycle can be delayed up to D56.

No dose adaptation is allowed during the DLT evaluation period (1 treatment cycle, 28 days of treatment administration).

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7.2 Supply Chain of IMPs

Packaging: The sponsor will supply the French commercial form, counter-labeled with the mentions of the research in accordance with Regulation (EU) No 536/2014 on clinical trials on medicinal products for human use.

Supplies: Treatment shipments will be made by the Clinical Trials Department (CTD) of AGEPS. Depending on the center's inclusion potential, an initial stock will be made available to the center or the treatments required for the research will be sent upon inclusion of a patient, based on the Dose Level.

Dispensing: Hospital pharmacists dispense the IMP in accordance with the specific research prescription completed and duly signed by the investigator by cycle of IMP. 2 dispensations will be needed by patient.

Returns and destruction: For expired treatments, destruction will be authorized by the sponsor's representative after monitoring. The becoming of products that are not expired at the end of the research will be determined on a case-by-case basis with the centers concerned.

7.3 Description of Additional medicinal product(s) (treatments required to conduct the study)

In accordance with Regulation (EU) No. 536/2014 of the European Parliament and of the Council of April 16, 2014, an auxiliary drug is a drug used for the purposes of a clinical trial in accordance with the protocol, but not as an investigational drug.

7.3.1 Auxiliary medicinal product: Posaconazole

Posaconazole is considered an auxiliary medicinal product as defined by regulatory guidelines, due to its systematic use as prophylaxis against invasive fungal infections. This is justified by the presence of profound neutropenia or impaired neutrophil function in patients with relapsed or refractory AML.

Posology : Loading dose of 300 mg (3 tablets of 100 mg) twice daily on the first day, then 300 mg (3 tablets of 100 mg) once daily on subsequent days. Each dose can be taken without regard to food intake. The duration of treatment is determined based on the recovery status of neutropenia or immune depression.

7.3.2 Auxiliary medicinal:

7.3.2.1 Prophylaxis of Tumour Lysis Syndrome

Hydration with isotonic fluids during Venetoclax exposure and prophylaxis with allopurinol/rasburicase are recommanded to prevent hyperuricemia may minimize potential complications of tumour lysis syndrome.

Following the end of cycle visit, patients will continue to be followed for treatment related TEAEs for 30 days following the end of actinomycin D exposure.

Patients will have regular post treatment visit until relapse or study exit after 12 months of follow-up.

The combination Actinomycin D and Venetoclax will not have additional treatment targeting acute myeloid leukemia.

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7.3.2.2 Management of prolonged cytopenia and delayed hematological recovery

Prolonged cytopenia is expected in patients with relapsed or refractory AML receiving antileukemic therapy. If hematologic recovery is delayed after completion of treatment, the following management rules apply.

If neutrophil recovery (ANC ≥ 0.5 × 10^9/L) and platelet recovery (≥ 50 × 10^9/L) have not occurred by Day 42 of cycles 1 and 2, a bone marrow examination should be performed to differentiate treatment-related marrow aplasia from persistent leukemia.

If the bone marrow shows persistent leukemia, the investigator may proceed with the next cycle of treatment according to clinical judgment.

If the bone marrow shows hypocellular marrow without evidence of active leukemia, treatment may be delayed until hematologic recovery occurs. The use of G-CSF after blast clearance may be considered at the investigator's discretion to facilitate neutrophil recovery.

Cycle 2 may be initiated up to Day 56 upon hematologic recovery.

If cytopenia persists beyond Day 56 despite absence of leukemia, permanent treatment discontinuation should be considered.

7.3.2.3 Management of infection-related complications

Patients with AML are at high risk of infectious complications due to disease-related and treatment-related immunosuppression.
Patients developing febrile neutropenia or suspected infection should be managed according to institutional and international guidelines, including:

- prompt evaluation and microbiological sampling
- initiation of empirical broad-spectrum intravenous antibiotics

Temporary treatment interruption may be required in case of severe infection or clinical instability, until adequate clinical improvement is achieved.

Supportive care, including antimicrobial therapy, transfusion support and intensive monitoring, should be provided according to institutional standards.

7.4 Authorised and prohibited treatments (medicinal, additional medicinal, surgical), including rescue medications

7.4.1 Authorised/Recommended treatments

- Patients should receive adequate IV and/or oral hydration during Venetoclax initiation.
- Patients should receive adequate anti-emetic prophylaxis during treatment, including setrons, anti-emetic neuroleptics, and/or substance P antagonists.
- In cases of blast clearance, investigators are allowed to use G-CSF at their discretion to promote neutrophil recovery, in accordance with institutional guidelines and standard clinical practice.

Patients should be hospitalized during the treatment at least until day 21 and/or achievement of a neutrophil count > 500/µL. Thereafter they can be discharged if, according to the judgment of

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the investigator, their medical condition allows it and when all non-hematological treatment-emergent AEs resolve to Grade ≤ 2.

7.4.2 Prohibited medications and medications with precautions for use

- Use of other anti-leukemic agents during the induction course, including other chemotherapy agents, midostaurine or other kinase inhibitors, is prohibited.
- Use of strong inhibitors and/or substrats of cytochrome 3A4 including not limited to, itraconazol, kétoconazol, voriconazol, clarithromycin, ritonavir, ciprofloxacin, diltiazem, érythromycin, fluconazole verapamil, terfenadine, astemizole, cisapride pimozide, halofantrine, or quinidine
- Use of inductors of cytochrome: carbamazepin, phenytoïn, rifampicin, bosentan, éfavirenz, étravirin, modafinil, nafcillin, St. John's Wort
- Use of thrombopoietin analogs is prohibited.
- Close monitoring is required for patients treated with immunosuppressors (such as cyclosporine and tacrolimus).

7.5 Methods for monitoring compliance with the treatment

The pharmacist is responsible for ensuring adequate storage conditions for experimental drugs and for guaranteeing the proper management of non-dispensed experimental drugs. This includes acknowledging the correct receipt of the treatments with each shipment (quantitatively and qualitatively). IMPs (Venetoclax and Actinomycin D) accounting will be set up in each center in order to trace:
- The date of receipt of the ME, the quantities, the batch numbers and the expiry dates of the ME,
- Per patient according to the patient's identification code: the date of dispensation, the quantities dispensed as well as their batch numbers and their expiry dates.
- Returns according to the date of receipt and the quantities of full/opened/empty boxes.

In application of the obligations of Good Clinical Practices, a patient card will be given to all patients and must specify the name, address and telephone number of the investigating center and starting date of treatment.

Study treatment will be administered in hospitalized patients, enabling to document perfectly the compliance of the treatment by nurses and clinical research assistants.

During hospitalization, the follow-up of the treatment is ensured by the investigative team in charge of the patient using eCRF and a notebook to ensure the traceability of the administrations.

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8 EFFICACY ASSESSMENT

8.1 Description of efficacy endpoints assessment parameters

Assessment of anti-leukemia efficacy:

- Complete response rate including CR, CRi (CR with incomplete hematologic recovery, meaning CR with platelet count <100,000/µL or absolute neutrophil count <1000/µL) and CRh (CR with partial hematologic recovery, meaning CR not fulfilling CR or CRi peripheral blood count criteria but with platelet count >50,000/µL AND absolute neutrophil count >500/µL).
- Morphologic leukemia-free state (MLFS): blastic criteria of response on the bone marrow aspirate, absence of extramedullary disease and no requirement of hematologic recovery.
- Partial response (PR): absence of circulating blasts; absence of extramedullary disease; decrease of bone marrow blast percentage between 5 and 25% and decrease of pretreatment bone marrow blast percentage by at least 50%
- Overall response rate (ORR) being defined as CR, CRi and CRh, MLFS and PR rates.
- Event-free survival (EFS); here events are defined as:
    - Non achievement of hematologic response including CR, CRi, CRh
    - Hematologic relapse or progressive disease
    - Initiation of any subsequent anti-leukemic therapy (excluding hydroxyurea)
    - Deaths prior to one of the events mentioned above

Reasons for subsequent therapy initiation (including intolerance, resistance, hematologic relapse, unsatisfactory MRD evolution), its type (intensive, less-intensive, best supportive care) and its frame (clinical trial, off-label or compassionate use) will be prospectively recorded.

- Progressive disease (PD):
    - >50% increase in marrow blasts over baseline (a minimum 15 point increase is required in cases with <30% blasts at baseline); or a persistent marrow blast percentage of >70% over at least 3 months, (without at least a 100% improvement in ANC to an absolute level >0.5x10^9/L and/or platelet count to > 50 x 10^9/L, non-transfused);
    - Or a new extramedullary disease

- Duration of response (DOR) defined as the date of response to the date of relapse up to 12 months,
- Relapse free survival (RFS) defined as the time from the date of inclusion to the date of relapse up to 12 months
- Overall survival (OS) defined as the time from the date of inclusion to the date of death (all-cause) up to 12 months
- Early mortality rates after 1 cycle of treatment (day 28-42) and after 2 cycles (day 56)
- Incidence of subsequent allogeneic HSCT, overall and in responding patients specifically
- Molecular Residual Disease (MRD) will be evaluated as follows: Peripheral blood RQ-PCR for NPM1 patients[16], BM RQ-PCR for t(8;21) and inv(16) patients[17], otherwise FCM.

Apart from specific mentions, all these hematological endpoints will be defined according to the last recommendation of the European Leukemia Network (ELN; currently the 2022 ELN recommendations in *Döhner H et al Blood 2022*).

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8.2 Anticipated methods and timetable for measuring, collecting and analysing the efficacy data

The endpoint of efficacy will be assessed at end of each cycle, between D28 and D42 at the latest.

Criteria of response are assessed from peripheral blood count and bone marrow aspirates performed according to this timeline.

- Morphological examination, will be performed in local hematology laboratories
- Eight color FCM MRD will be performed on bone marrow aspirates sampled according to the same timeline.

In *NPM1* mutated patients, standard PCR-based MRD will also be conducted on the bone marrow and peripheral blood to guide post-remission therapies.

Samples for biomarkers and PD (30 ml of blood and 3 ml of bone marrow aspirates in EDTA tubes) should be sent at room temperature within 24 hours to the Laboratoire d'Hématologie Hôpital Saint Louis 1 avenue Claude Vellefaux 75010 Paris.

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9 SPECIFIC STUDY COMMITTEES

Two committees will govern the study:

English name	French name
Data Safety Monitoring Board (DSMB)	Comité de surveillance indépendant (CSI)
Steering Committee	Comité de Pilotage

9.1 Steering Committee

9.1.1 Role

The role of the steering committee is manifold:

- Oversee the progression of the study
- Propose procedures to be followed during the study, acknowledging any recommendations from the DSMB. The DRCI sponsor retains decision-making authority.
- Review and validate the analysis plan of the data.
- Review and validate the draft of the publications.

9.1.2 Composition

The steering committee should meet the main players of the study:

- The coordinator investigator (Dr Sofiane Fodil)
- The biostatistician in charge of the project (Prof Lucie Biard, Saint-Louis, Unité de Recherche Clinique, Department of Biostatistics)
- The pharmacologist (Dr Lauriane Goldwirt, Saint Louis pharmacology department, Paris)
- The principal investigators of each participating centres
- The scientific director of the study (Prof Raphaël Itzykson)
- A representative of the sponsor (Petra Slivova)
- A representative of the safety department (Valentin Hosansky)
- A representative of the pharmaceutical coordination (Robin Charreteur)
- The coordination team of THEMA (Karine Celli-Lebras, Dr Renaud Buffet, Dr Laure Gilles).

9.2 DSMB

See the section 10.3.4.

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10 SAFETY ASSESSMENT - RISKS AND BURDEN ADDED BY THE STUDY

10.1 Description of Safety endpoints assessment parameters

Safety endpoints including DLT definition are described in section **4.1** of the protocol.

Identification and grading of AEs is entirely based on NCI CTCAE version 6.0 of the NCI.

10.2 Anticipated methods and timetable for measuring, collecting and analysing the safety endpoints

Safety endpoints are easily monitored through clinical examination and usual laboratory tests,

T**he eCRF should be filled in nearly real time** and monitored as soon as possible after the end of cycle visit. Real-time assessment of DLTs is central to the Bayesian continual reassessment method (CRM). All monitored DLTs will inform the modelling of the DL allocated to the next enrolled patient (see details on enrollment procedure in **Section 5.1.2** and on the CRM model in **Section 12**).

10.3 Recording and reporting adverse events

10.3.1 Definitions

- **Adverse event (AE)**

Any untoward medical occurrence in a subject, to whom a medicinal product is administered and which does not necessarily have a causal relationship with this treatment.

- **Serious adverse event (SAE)**

Any untoward medical occurrence that at any dose requires inpatient hospitalisation or prolongation of existing hospitalisation, results in persistent or significant disability or incapacity, results in a congenital anomaly or birth defect, is life-threatening, or results in death.
Certain medical events may jeopardise the subject or may require an intervention to prevent a SAE, known as "important medical events" should also be considered as serious adverse events.

- **Suspected Unexpected serious adverse reaction (SUSAR)**

A serious adverse reaction, the nature, severity or outcome of which is not consistent with the reference safety information.

- **Unexpected event**

An unexpected event which affects the benefit-risk balance of the clinical trial, but are not suspected unexpected serious adverse reactions as referred to in Article 42 of Regulation (EU) No 536/2014 of the European parliament. That notification shall be made without undue delay but no later than 15 days from the date the sponsor became aware of this event.

- **Urgent safety measure**

Where an unexpected event is likely to seriously affect the benefit-risk balance, the sponsor and the investigator shall take appropriate urgent safety measures to protect the subjects.
The sponsor shall notify the Member States concerned, through the EU portal, of the event and the measures taken.
That notification shall be made without undue delay but no later than seven days from the date the measures have been taken.

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10.3.2 The role of the investigator

The investigator must **assess the seriousness of each adverse event** and record all serious and non-serious adverse events, and all special circumstances (medication errors, in utero exposure / pregnancy / breastfeeding, uses outside what is foreseen in the protocol, including misuse and abuse of the product) in the case report form (eCRF).

The investigator must **document** serious adverse events **as thoroughly as possible** and provide a definitive medical diagnosis, if possible.

The investigator must **assess the intensity** of the adverse events: by using the Common Terminology Criteria for Adverse Events (CTCAE of the National Cancer Institute), version 6.0 of 22 July 2025.

The investigator must **assess the causal relationship** between serious adverse events and:

- the investigational medicinal product(s) (IMP): venetoclax and actinomycin D,
- and/or interventions/procedures added by the study (blood sample and bone marrow aspiration),
- and/or auxiliary medicinal product (posaconazole),
- and/or interaction between auxiliary medicinal product and the investigational medicinal products.

In case of adverse reactions only related to authorised auxiliary medicinal product(s) in accordance with Regulation (EC) No 726/2004 or in any Member State concerned in accordance with Directive 2001/83/EC: SAE must be notified to sponsor's safety department as soon as possible and also the investigator must report to the relevant regional pharmacovigilance center (Centre Régional de Pharmacovigilance (CRPV)) or to www.signalement-sante.gouv.fr.
In case of a suspected interaction between authorised auxiliary medicinal product(s) and the investigational medicinal product(s), the reporting rules for the IMP apply: SAE must be notified to the sponsor's safety department as soon as possible.

The method to be used by the investigator is based on 2 causality terms (EudraVigilance Clinical Trial Module (EVCTM) method):

- Related
- Not related

10.3.2.1 Serious adverse events that require the investigator to notify the sponsor without delay

The investigator notifies the sponsor without undue delay but not later than within 24 hours on the day the investigator becomes aware of any serious adverse event which occurs during a study that meets the description in article 41 of Regulation (EU) N°536/2014, with the exception of any event which is listed in the protocol and, if applicable, in the investigator's brochure as not requiring notification.

A serious adverse event is any untoward medical occurrence that:

1- results in death
2- is life-threatening to the participant enrolled in the study
3- requires hospitalisation or prolongation of existing hospitalisation
4- results in persistent or significant disability/incapacity
5- is a congenital anomaly/birth defect
6- important medical event*

***** *Certain medical events may jeopardise the subject or may require an intervention to prevent a SAE, known as "important medical events" should also be considered as serious adverse events.*

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Moreover, special circumstances require the investigator to notify the sponsor without delay are:
- medication errors,
- in utero exposure / pregnancy / breastfeeding
- uses outside what is foreseen in the protocol, including misuse and abuse of the product.

All pregnancies must be reported to the safety department by the investigators within 24 hours on the day the investigator becomes aware of it.

Only special circumstances associated to an SAE must be reported to the safety department by the investigators and must be recorded in the "SAE" module of the CRF.

Special circumstances not associated to an SAE must be reported to the CRF in the "AE" module by the investigators and notify as a deviation to the protocol by the clinical trial unit.

10.3.2.2 Specific features of the protocol

10.3.2.2.1 Other events that require the investigator to notify without delay the sponsor

- Adverse events particularly followed by the sponsor for the safety and assessment

Any grade ≥ 3 non hematological adverse events fitting with DLT criteria with the exception of biological adverse event without clinical signs associated.

- Death of any cause (with the exception of deaths caused by clear evidence of disease progression, defined as a leukocytosis greater than 50 × 10^9/L with no evidence of toxicity)
- Prolonged myelosuppression on day 42 from start of therapy or later without evidence of leukemia defined by the absence of platelet recovery above 50 × 10^9/L without transfusion support within the last week and the absence of neutrophil recovery with ANC > 1 × 10^9/L
- Grade ≥ 3 veno-occlusive disease (VOD)
- Grade ≥ 3 non- hematological toxicity with the exception of:
  - Grade 3 or 4 infections
  - Grade 3 or 4 fever with neutropenia
  - Grade 3 nausea, vomiting or diarrhea that can be managed to ≤ Grade 2 within 72 hours of symptomatic treatment
  - Grade 3 mucositis or stomatitis that can be managed to ≤ Grade 2 within 72 hours of symptomatic treatment
  - Grade 3 tumor lysis syndrome that resolves within 72 hours of onset with medical treatment
  - Grade 3 or 4 hemorrhage

These events must be notified by the investigator to the sponsor without delay on the day the investigator becomes aware of these adverse events, in the same manner and within the same deadline as for serious adverse events (see above).

For these adverse events particularly followed by the sponsor, please, check the "important medical event" box as a seriousness criterion in the SAE notification form.

- In utero exposure

The investigator must notify the sponsor without delay on the day the investigator becomes aware of any pregnancy that occurs during the study, even if it is not associated with an adverse event.

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If the investigational medicinal product is genotoxic, every case of maternal or paternal exposure must be notified.

- Exposure while breastfeeding

Exposure while breastfeeding occurs if an infant or a child may have been exposed to a medicinal product *via* the breast milk of a mother being treated with an investigational medicinal product.

Even if it is not associated with an adverse event, the investigator must notify the sponsor without delay on the day the investigator becomes aware of the exposure while breastfeeding.

10.3.2.2.2 Serious adverse events that do not require the investigator to notify the sponsor without delay

- An extraction of all these SAE from the e-CRF will be carried out by Clinical Trial Unit (including randomization group if applicable, seriousness criteria, investigator's assessment and SAE grade if applicable) and sent by email to the Safety Department and to the members of the Data Safety Monitoring Board according to a frequency defined in the DSMB charter:

- Every 6 patients included in the study.
- Every 6 months after the first inclusion

Whatever event occurs first.

These extractions will be sent to the DRCI safety department at the following address: expertisecsi.drc@aphp.fr

- Hospitalization for the treatment of AML (without deterioration of the patient's health condition).
- Hematological toxicities except if persisting at grade ≥3 after day 42 in the absence of proven disease persistence by bone marrow aspiration and/or biopsy. This rule is justified by the fact that hematological recovery after effective anti-leukemia treatment is observed in median at day 22, with 17% of patients still having grade 4 neutropenia by day 28[23].

- Other circumstances that do not require the investigator to notify the sponsor without delay
  - Hospitalisation for a pre-existing illness or condition without deterioration
  - Hospitalisation for a medical or surgical treatment scheduled prior to the study
  - Admission for social or administrative reasons
  - Medication errors or uses outside what is foreseen in the protocol, including misuse and abuse of the product without SAE and not in relationship with a serious breach

10.3.2.3 Period during which SAEs must be notified without delay by the investigator to the sponsor

The investigator must notify the sponsor without delay of any serious adverse events as defined in the corresponding section:

- from the date on which the participant begins treatment.
- until 4 weeks after the end of treatment **except fatal event** (which are to notify throughout the whole follow-up period required)
- indefinitely, if the SAE is likely to be due to the investigational medicinal product or to the study interventions (e.g. serious reactions that could appear long after exposure to the medication, such as cancers or congenital abnormalities) *.

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*NB: In this last case, the investigator **does not have to collect indefinitely in the case report form (CRF or eCRF)** all **SAEs possibly related** to the clinical trial, **but must notify them**, to the sponsor, **as soon as he/she becomes aware of them, by the SAE notification form or by email or by fax (as described below).**

10.3.2.4 Procedures and deadlines for notifying the sponsor

The initial notification of an SAE must be provided in a written report signed by the investigator using a SAE notification form specific to the study and intended for this purpose (in the case report form).

Each item in this form must be completed by the investigator so that the sponsor can carry out the appropriate analysis.

The initial notification to the sponsor of a serious adverse event must be quickly followed by an additional detailed written report (or reports) so that the case outcome may be monitored by the Safety Department or to provide further information.

Whenever possible, the investigator will provide the sponsor with any documents that may be useful (medical reports, laboratory test results, results from additional examinations, etc.). These documents must be non-identifying. In addition, the documents must include the following: study acronym and participant number.

Any adverse event will be monitored until fully resolved (stabilisation at a level considered acceptable by the investigator, or return to the previous state) even if the participant has left the study.

The initial notification, the SAE follow-up reports and all other documents must be sent to the sponsor's Safety Department by email (eig-vigilance.drc@aphp.fr). It should be noted that it is possible to send SAE reports to the Safety Department by fax to +33 (0)1 44 84 17 99 only in the event of a failed attempt to send the SAE report by email (in order to avoid duplication).

When sending the email, please:

- Adopt a standardized nomming of the email subject in the following form:

<< Objet : YYYYYY_XXXXXX_jjmmaaaa (avec YYYYYY : code de la recherche, XXXXXX : acronyme de la recherche et jjmmaaaa : date de transmission). >>

- Send a SAE initial notification form and/or a follow-up report concerning a single participant for a given SAE, attachment may contain one (or more) document(s) (follow-up and hospitalization reports, for example).
The total size of the email must be less than 8 MB, otherwise please send several e-mails.

- Ensure that all documents transmitted (e.g. hospital reports) are anonymized and identified with the participant's identification number.

For studies which use e-CRFs

- the investigator completes the SAE notification form in the e-CRF, then validates, prints, and signs the form before sending it by email;
- if it is not possible to connect to the e-CRF, the investigator should complete, sign, and send the SAE notification form to the Safety Department. As soon as the connection is restored, the SAE notification form in the e-CRF must be duly completed.

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The investigator must respond to all requests from the sponsor for additional information (these new informations must be also recorded in the e-CRF).
For all questions relating to the notification of an adverse event, the Safety Department can be contacted via email: vigilance.drc@aphp.fr.

For cases of *in utero* exposure, the investigator will complete the Notification and Follow-up form for a pregnancy occurring during participation in a study".
The investigator must monitor the pregnant woman throughout her pregnancy or until the pregnancy is terminated, and must notify the sponsor of the outcome of the pregnancy using this form.
If the outcome of the pregnancy falls within the definition of a serious adverse event (miscarriage, pregnancy termination, foetal death, congenital abnormality, etc.), the investigator must follow the procedure for reporting SAEs.
The initial pregnancy notification, the SAE follow-up reports, and any other documents will be sent to the sponsor according to the same procedures specified herein.

10.3.3 Role of the sponsor

The sponsor, represented by its Safety Department, shall continuously, throughout the trial, assess the safety of each investigational medicinal product throughout the study.

10.3.3.1 Analysis and declaration of serious adverse events

The sponsor assesses:

- the **seriousness** of all the adverse events reported
- the **causal relationship** between these events and each investigational medicinal product and/or the auxiliary medicinal products and/or interaction between the investigational medicinal products and the auxiliary medicinal products, and any concomitant treatments,
  All serious adverse events which the investigator and/or the sponsor reasonably believe may have a causal relationship with the investigational medicinal product are classed as suspected serious adverse reactions.
- the **expected or unexpected nature** of the serious adverse reactions related to IMP and to the interventions, procedures or examinations specific to the study.
  Any serious adverse reaction whose nature, severity, frequency or outcome is inconsistent with the safety information described in the summary of product characteristics, is considered unexpected.
  The sponsor, represented by its Safety Department, assesses the expected/unexpected nature of a serious adverse reaction based on the information described below.

- For serious adverse events likely to be related to investigational medicinal products: refer to the section 4.8 “Undesirable effects” of the SmPC of venetoclax and/or Actinomycin D (cf. Appendix 18.4) enclosed in CTIS platform and in the investigator study file.

❖ For serious adverse events that may be related to the auxiliary medicinal product: refer to the SmPC for posaconazole mentioned in the cover letter, filed in CTIS platform.

❖ The expected serious adverse events potentially related to the interventions, procedures or examinations specific to the study are:

- For blood samples: hematoma, pain, vagal malaise, mild bleeding
- For bone marrow aspiration: bleeding, pain, hematoma

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The sponsor will report all suspected unexpected serious adverse reactions (SUSARs) via Eudravigilance, within the regulatory time frame, to the competent authority:

- in the case of fatal or life-threatening suspected unexpected serious adverse reactions, as soon as possible and in any event not later than 7 days after the sponsor became aware of the reaction;
- in the case of non-fatal or non-life-threatening suspected unexpected serious adverse reactions, not later than 15 days after the sponsor became aware of the reaction;
- in the case of a suspected unexpected serious adverse reaction which was initially considered to be non-fatal or nonlife threatening but which turns out to be fatal or life-threatening, as soon as possible and in any event not later than 7 days after the sponsor became aware of the reaction being fatal or life-threatening
- in the uncompleted case of fatal or life-threatening suspected unexpected serious adverse reactions, all additional information to complete this case have to be reported as soon as possible and in any event not later than 8 days after the initial report

All additional, relevant information must be declared by the sponsor in the form of follow-up reports within a period of 15 days starting from when the sponsor had this information.

The sponsor must notify all the investigators involved about any information that could adversely affect the safety of the research participants.

10.3.3.2 Analysis and declaration of other safety data

This means any new information that prompts a reassessment of the risk/benefit ratio of the trial or of the product under investigation, or which could be sufficient to consider changes to the use of the product, the trial procedures or trial documentation, or the suspension, cancellation or amendment of the research trial or of other similar trials.

The sponsor will report in CTIS platform any emerging safety issue and, if applicable, describe what measures have been taken.

The unexpected event shall be notified without undue delay but no later than 15 days from the date the sponsor became aware of this event.

The urgent safety measures shall be notified without undue delay but no later than 7 days from the measures have been taken.

Following the initial declaration of any emerging safety issue, the sponsor will address to competent authorities any additional relevant information about the emerging safety issue in a follow-up report.

10.3.3.3 Annual safety report

Once a year for the duration of the clinical study, the sponsor must produce an annual safety report (Development Safety Update Report - DSUR) which includes, in particular:

- a safety analysis for the research participants,
- a description of the patients included in the study (demographic profile etc.)
- a list of all the suspected serious adverse reactions that occurred during the period covered by the report,
- summary tables of all the serious adverse events that have occurred since the start of the study,

The sponsor produces one annual safety report (Development Safety Update Report - DSUR) for one clinical trial.

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The report must be submitted in CTIS no later than 60 days after the anniversary of the date on which the competent authority authorised the trial.

The final annual safety report must be submitted in CTIS no later than 60 days after the end date of the clinical trial.

The end date of the clinical trial is defined as the date of the last visit of the last subject.

10.3.4 Data Safety Monitoring Board (DSMB)

A Data Safety Monitoring Board (DSMB) will be established for this trial. The DSMB must hold its first meeting before the first participant is enrolled and ideally before the protocol is submitted to the competent authority and the CPP (Research Ethics Committee).

The DSMB members are (to be confirmed):

- Two experts in AML:
    - One national expert : Corentin Orvain
    - One international expert : Andrius Zucenka
- One vigilance expert : Marine Rousset

The DSMB will operate in accordance with the sponsor's procedures. The DSMB works in an advisory capacity only and the sponsor retains all decision-making authority.

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11 DATA MANAGEMENT

11.1 Data access

In accordance with GCPs:

- the sponsor is responsible for ensuring all parties involved in the study agree to guarantee direct access to all locations where the study will be carried out, the source data, the source documents and the reports, for the purposes of the sponsor's quality control and audit procedures or inspections by the competent authority
- the investigators will ensure the persons in charge of monitoring, quality control and auditing or inspecting the interventional study involving human participants have access to the documents and personal data strictly necessary for these tasks, in accordance with the statutory and regulatory provisions in force (Articles L.1121-3 and R.5121-13 of the French Public Health Code)

11.2 Source documents

Source documents are defined as any original document or item that can prove the existence or accuracy of a data or a fact recorded during the study. These documents will be kept in accordance with the regulations in force by the investigator or by the hospital in the case of a hospital medical file.

In the context of the study source document are medical files, original biological examination results, summary from imaging examinations.

11.3 Data confidentiality

The persons responsible for the quality control of clinical studies (Article L.1121-3 of the *Code de la Santé Publique* [French Public Health Code]) will take all necessary precautions to ensure the confidentiality of information relating to the investigational medicinal products, the study, the study participants and in particular their identity and the results obtained.

These persons, as well as the investigators themselves, are bound by professional secrecy (in accordance with the conditions set out in Articles 226-13 and 226-14 of the *Code Pénal* [French Criminal Code]).

During and after the research involving human participants, all data collected concerning the participants and sent to the sponsor by the investigators (or any other specialised collaborators) will be rendered non-identifying.

Under no circumstances shall the names and addresses of the participants involved be shown. Only the participant’s initials will be recorded, accompanied by an encoded number specific to the study indicating the order of enrolment.

The sponsor will ensure that each participant has given written permission for any personal information about him or her which is strictly necessary for the quality control of the study to be accessed.

11.4 Data processing and storage of research documents and data

11.4.1 Identification of the data processing manager and location(s)

Prof. Lucie BIARD from the Clinical Research Unit (URC, an INCa labeled Data Center) and Service Biostatistique et Informatique Médicale (SBIM) of Hôpital Saint Louis, AP-HP, Paris will

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be responsible for data management and the relevant procedures. The same goes for conducting the statistical analysis.

11.4.2 Data entry

Non-identifying data will be entered electronically via a web browser.

11.5 Data ownership

AP-HP is the owner of the data. The data cannot be used or disclosed to a third party without its prior permission.

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12 STATISTICAL ASPECTS

12.1 Description of statistical methods to be used including the timetable for the planned interim analyses

As a general strategy, continuous variables will be summarized using summary measures (median and interquartile range). Frequency distributions (counts and percentages) will be used to summarize categorical endpoints. Similarly, characteristics of patients will be presented using summary measures (median and interquartile range for quantitative characteristics and counts and percentages for categorical characteristics). Time-to-event endpoints will be summarized using Kaplan Meier estimates. Such descriptive analyses will be performed overall and according to the dose level group.

Primary analysis: The primary estimand will be the dose-limiting toxicity rate at the end of the first cycle of treatment (28 to 42 days) at each of the candidate dose level (DL1 to 3), estimated with the Bayesian posterior mean probability, in patients with in relapsed/refractory acute myeloid leukemia receiving the combination Actinomycin D/Venetoclax at any of the candidate dose levels. The primary endpoint will be the presence of at least oneDLT.. Dose finding analyses will be performed on the modified intent-to-treat set (mITT, see section 12.7) according to the dose level effectively received as first treatment cycle (1 to 3 days of Actinomycine), in the case of intercurrent initial dose adjustment decision. Otherwise, in the case of intercurrent dose adjustment during the second cycle, patients will remain analyzed in their initial dose group. Observations with intercurrent events such as treatment discontinuation or death will remain included in the analyses, within the time-to-event modeling framework[29].

A Bayesian adaptive dose-escalation phase I design will be used for the trial primary objective. It allows sequential monitoring of data during the trial, using all current evidence for straightforward inference on safety and decision making to select and evaluate the maximum tolerated dose (MTD). Af one-stage Bayesian time-to-event continual reassessment method (TITE-CRM) will be used for up to 36 patients, with fixed cohort size of 3[24] for dose assignments (minimum 3 patients per dose level, maximum 36, given safety rules described below). The sample size will likely allow convergence of dose assignments on the estimated MTD and result in an expanded sub-sample of patients assigned to the estimated MTD. Target maximum DLT rate for the definition of the MTD will be 33%. An empirical working model will be used. Analysis of the DLT probability will be unadjusted, and sequential, upon initiation of any new dose-finding cohort.

The initial dose will be Dose Level 1 (DL1); dose assignments will be performed by cohorts of 3 patients. The CRM will be calibrated prior to enrolment initiation, using the indifference interval approach[25,26] to obtain a skeleton on the prior DLT rate assuming DL2 is the anticipated MTD, and accounting for safety rules (see below), with evaluation of operating characteristics by simulations. The design skeleton will be defined prior to inclusion of the first patient and included on the protocol and the SAP[27].

Safety rules will be implemented, including:

- No dose skipping will be allowed during dose escalation;
- For each newly escalated dose level, the first three patients treated at this dose level will be included at least one week apart (staggered accrual);
- The trial will be terminated if the posterior probability of DLT within the primary observation window at dose level 1 (DL1) is greater than 95%.

Sensitivity analyses on the complete dose escalation data at the end of the trial, and including the second cycle of combination Venetoclax + Actinomycin, may be performed accounting for

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the dose levels effectively at each cycle of treatment, using models allowing intra-patient dose-adaptation[30, 31, 31]. Such analyses will be detailed in the SAP.

12.2 Calculation hypotheses for the number of participants required and the result

A one-stage Bayesian time-to-event continual reassessment method (TITE-CRM) will be used with cohort size of 3[24]. A maximum sample size of 36 ensures a probability of correct select of 0.6 for the CRM, assuming a 1.5 odds ratio for the true logistic dose-toxicity curve and using a basic arbitrary skeleton with DL2 as the anticipated MTD[28]. The skeleton and prior variance of the dose-toxicity model parameter will be further calibrated by simulation[25-26], to optimize the probability of correct selection (PCS), accounting for the trial design and additional safety rules (see section 12.1), and detailed in the SAP before trial initiation. The sample size will likely allow convergence of dose assignments on the estimated MTD and result in an expanded sub-sample of patients assigned to the estimated MTD.

12.3 Anticipated level of statistical significance

No hypothesis testing will be performed for the primary analyses.

12.4 Statistical criteria for termination of the study.

A safety stopping rule will be implemented: The trial will be terminated if the posterior probability of DLT within the primary observation window at dose level 1 (DL1) is greater than 95%.

12.5 Method for taking into account missing, unused or invalid data

Missing data will be described, overall and by dose level.

Missing data on the primary endpoint is unlikely (toxicity endpoint in a phase I trial). Nevertheless, any incomplete data on the primary endpoint will be straightforwardly handled by the time-to-event design, censoring observation at the last available measure[29].

For the secondary endpoints, analysis will be firstly performed on the modified intent to treat set (see section 12.7). Missing values on endpoints will be handled by simple imputation (for continuous endpoints: last observation carried forward when available or mean value otherwise, for binary endpoints: failure when appropriate). Sensitivity analyses will be performed on the complete case set and upon various hypotheses on missing values.

12.6 Management of modifications made to the analysis plan for the initial strategy.

A statistical analysis plan will be defined including a detailed and technical description of statistical methods. As recommended in ICH E9 Sections 5.1 et 5.2 [Lewis JA 1999]. All modifications of the initial plan will be tracked and recorded in the statistical analysis plan, and submitted to the scientific committee and the sponsor.

12.7 Choice of individuals to be included in the analyses

Study samples:

- Intention to treat sample (ITT): refers to all included subjects, in the dose level group as assigned.
- Modified Intention to treat sample (mITT): refers to the ITT patients who received at least one administration of the experimental treatment, analyzed in the group of the dose level they actually received in the first treatment cycle (1 to 3 days of Actinomycine). This will refer to the primary analyses

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- Per protocol sample: Includes all subjects from the mITT set without any major violations from the assigned dose schedule (cycles 1 and 2), which could affect the evaluation of the primary efficacy endpoint. This will be used as sensitivity analyses.
- Complete case set: for a given variable, refers to the set of patients without missing data on this variable. Will be used for sensitivity analyses.

Primary analyses will be performed on the mITT sample, with description of baseline characteristic and of trial endpoints overall and by dose level, following guidelines for dose-finding trials[27].

Subgroup analyses according to MRD status and HSCT status will be performed for the secondary endpoints.

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13 QUALITY CONTROL AND ASSURANCE

Every research project involving human participants managed by AP-HP is ranked according to the projected risk incurred by study participants using the classification system specific to AP-HP sponsored interventional research involving human participants.

13.1 General organisation

The sponsor must ensure the safety and respect of individuals who have agreed to participate in the study. The sponsor must implement a quality assurance system to best monitor the implementation of the study in the investigation centers.

For this purpose, the sponsor shall assign Clinical Research Associates (CRA) whose primary role is to carry out regular follow-up visits at the study locations, after having carried out the initial visits.

The purpose of monitoring the study, as defined in Good Clinical Practices (GCP section 5.18.1), is to verify that:

- the rights, safety and protection of the research participants are met
- the data reported are exact, complete and consistent with the source documents
- the study is carried out in accordance with the protocol in force, the GCPs and the statutory and regulatory provisions in force

13.1.1 Strategy for centre opening

The strategy for opening the centers established for this study is determined using the appropriate monitoring plan.

13.1.2 Scope of centre monitoring

In the case of this level D risk study the appropriate monitoring level has been determined based on the complexity, the impact and the budget for the study. Therefore, in agreement with the coordinating investigator, the sponsor has determined the logistical score and impact, resulting in a study monitoring level to be implemented: level D.

13.2 Quality control

A Clinical Research Associate (CRA) appointed by the sponsor will be responsible for the good completion of the study, for collecting, documenting, recording and reporting all handwritten data, in accordance with the Standard Operating Procedures applied within the DRCI (Clinical Research and Innovation Department) and in accordance with Good Clinical Practices as well as the statutory and regulatory requirements.

The investigator and the members of the investigator's team agree to make themselves available during regular Quality Control visits carried out by the Clinical Research Associate. During these visits, the following elements will be reviewed depending on the monitoring level:

- written consent,
- compliance with the study protocol and with the procedures defined therein,
- quality of the data collected in the case report form: accuracy, missing data, consistency of the data with the "source" documents (medical files, appointment books, original copies of laboratory results, etc.),
- management of the treatments used.

13.3 Case report forms

Electronic CRF:

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All information required by the protocol must be entered in the case report forms. The data must be collected as and when they are obtained, and clearly recorded in these case report forms. Any missing data must be coded.
Every site will have access to the electronic case report forms via a web-based data collection system. Investigators will be given a document offering guidance on using this tool.

When the investigators complete the case report form via the Internet, the CRA can view the data quickly and remotely. The investigator is responsible for the accuracy, quality and relevance of all the data entered. In addition, the data are immediately verified as they are entered, thanks to consistency checks. To this end, the investigator must validate any changes to the values in the case report form. An audit trail will be kept of all changes. A justification can be added when applicable, as a comment.

A print-out, authenticated (signed and dated) by the investigator, will be requested at the end of the study. The investigator must archive a copy of the authenticated document that was issued to the sponsor.

13.4 Management of non-compliances

The investigators agree to consent to the quality assurance audits carried out by the sponsor as well as the inspections carried out by the competent authorities. All data, documents and reports may be subject to regulatory audits and inspections. These audits and inspections cannot be refused on the grounds of medical secrecy.

An audit can be carried out at any time by individuals appointed by the sponsor and independent of those responsible for the research. The aim of the audits is to ensure the quality of the study, the validity of the results and compliance with the legislation and regulations in force.
The persons who manage and monitor the study agree to comply with the sponsor's requirements and with the competent authority regarding study audits or inspections.

The audit may encompass all stages of the study, from the development of the protocol to the publication of the results, including the storage of the data used or produced as part of the study.

13.5 Principal Investigator's commitment to assume responsibility

Before starting the study, each investigator will give the sponsor's representative a copy of his/her updated personal *curriculum vitæ*, signed and dated less than one year, with his/her RPPS number (*Répertoire Partagé des Professionnels de Santé*, Collective Database of Health Professionals). The CV must include any previous involvement in clinical research and related training.

Each investigator will commit to comply with legislation and to conduct the study in line with GCP, in accordance with the Declaration of Helsinki.

The Principal Investigator at each participating site will sign a commitment of responsibility (standard DRCI document) which will be sent to the sponsor's representative.

The investigators and their staff will sign a delegation of duties form specifying each person's role and will provide their CVs.

13.6 Pharmacist's commitment of responsibility

Production problems must be recorded according to the Manufacturing unit (LTCG) procedures.

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14 ETHICAL AND LEGAL CONSIDERATIONS

14.1 Methods for informing research participants and obtaining their consent

In accordance with Article L1122-1-1 of the Code de la Santé Publique (French Public Health Code), no interventional research involving human participants can be carried out on a person without his/her freely given and informed consent, obtained in writing after the person has been given the information specified in Article L.1122-1 of the aforementioned Code.

A minimum reflection period of ***72 hours*** is given to the individual between the time when he or she is informed and when he or she signs the consent form.

The person's freely given, written, informed consent will be obtained by the principal investigator or a physician representing the investigator before the person is enrolled in the study.

A copy of the information note and consent form, signed and dated by the research participant and by the principal investigator or the physician representing the investigator will be given to the individual prior to their participation in the study. The principal investigator or the physician representing him/her will keep a copy.

At the end of the study, one copy will be placed in a tamper-proof sealed envelope containing all the consent forms. This envelope will be archived by the sponsor.

In addition, the investigator will specify in the person's medical file the person's participation in the research, the procedures for obtaining his/her consent as well as the methods used for providing information for the purpose of collecting it. The investigator will retain one copy of the signed and dated consent form.

14.1 Compensation for participants

No compensation for participants is planned.

14.2 Authorization for the research location

The study will be carried out in treatment units on individuals presenting a clinical condition for which the units are specialized and who require interventions that are routinely performed at those units. Therefore, the research location does not require specific authorization.

14.3 Legal obligations

14.3.1 Role of the sponsor

Assistance Publique - Hôpitaux de Paris (AP-HP) is the sponsor of this study and, by delegation, the DRCI (Clinical Research and Innovation Department) carries out the study's missions in accordance with Article L.1121-1 of the *Code de la Santé Publique (French Public Health Code)*. Assistance Publique - Hôpitaux de Paris reserves the right to halt the study at any time for medical or administrative reasons. In this case, notification will be sent to the investigator.

14.3.2 Request for approval from the CPP (Research Ethics Committee)

Prior to starting the study, AP-HP, as sponsor, must obtain for this interventional study involving human participants concerning a medicinal product for human use, approval from the appropriate CPP (Research Ethics Committee), within the scope of its authority and in accordance with in force legislation and regulatory requirements.

14.3.3 Request for authorization from ANSM

Prior to starting the study, AP-HP, as sponsor, must obtains authorization from the ANSM (French Health Products Safety Agency) for the interventional study involving human participants concerning a medicinal product for human use, within the scope of the ANSM's authority and in accordance with in force legislation and regulatory requirements.

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14.3.4 Procedures relating to data protection regulations

The computer file used for this research is implemented in accordance with French (amended "Informatique et Libertés" law governing data protection) and European (General Data Protection Regulation – GDPR) regulations.

The research is governed by the "Reference Methodology" (MR-001)

- Commitment to comply with "Reference Methodology" MR-001

This research is governed by the CNIL (French Data Protection Agency) "Reference Methodology for processing personal data used within the scope of health research" (amended MR-001). AP-HP, as sponsor of the research, has signed a declaration of compliance with this "Reference Methodology".

14.3.5 Amendments to the research

Any substantial modification to the protocol by the coordinating investigator must be sent to the sponsor for approval. After approval is given, the sponsor must obtain, approval from the CPP (Research Ethics Committee) and authorization from the ANSM within the scope of their respective authorities, before the amendment can be implemented.

The information note and the consent form can be revised if necessary, in particular in case of a substantial amendment to the study or if adverse reactions occur.

14.3.6 Final study report

The final report for the research involving human participants referred to in Article R1123-67 of the *Code de la Santé Publique* (French Public Health Code) is written and signed by the sponsor and the investigator. A report summary drafted according to the reference plan of the competent authority must be sent to the competent authority within a period of one year following the end of the study, i.e., the end of the participation of the last participant in the study.

14.3.7 Archiving

Specific documents for an interventional study involving human participants concerning a medicinal product for human use will be archived by the investigator and the sponsor for 15 years after the end of the research.

This indexed archiving includes, in particular:

- A sealed envelope for the investigator containing **a copy of all the information notes and consent forms signed by all individuals at the center who participated in the study**;
- A sealed envelope for the sponsor, containing **a copy of all the information notes and consent forms signed by all individuals at the center who participated in the study**;
- "Study" binders for the Investigator and the sponsor, including (non-exhaustive list):
  - the successive versions of the protocol (identified by the version number and its date), and any appendices
  - the ANSM authorizations and CPP (Research Ethics Committee) decisions
  - any correspondence
  - the enrolment list or register
  - the appendices specific to the research
  - final study report
- The data collection documents

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15 FUNDING AND INSURANCE

The study is funded by the PHRC-I. A co funding has been granted by L'Institut de la Leucémie (IHU THEMA). The resources of the THEMA program (study coordination) will contribute to the study conduction.

15.1 Insurance

For the duration of the study, the Sponsor will take out an insurance policy covering the sponsor's own public liability, as well as the public liability for all the physicians involved in the study. The sponsor will also provide full compensation for any damages caused by the study to the participant enrolled and their beneficiaries, unless the sponsor can prove that the harm is not the fault of the sponsor or any collaborator. Compensation cannot be refused on the grounds of a third-party act or the voluntary withdrawal of the person who initially consented to participate in the study.

Assistance Publique - Hôpitaux de Paris (AP-HP) has taken out insurance with HDI-GLOBAL SE through the insurance broker BIOMEDIC-INSURE for the full study period, which covers its own public liability and that of any collaborator (physician or research staff), in accordance with Article L.1121-10 of the *Code de la Santé Publique* (French Public Health Code).

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16 PUBLICATION RULES

The author(s) of any publication relating to this study must include the AP-HP among their affiliations and must name the sponsor AP-HP (DRCI) and the source of funding,

16.1 Mention of AP-HP affiliation for projects sponsored by AP-HP

- If an author has several affiliations, the order in which the institutions are mentioned (AP-HP, University, INSERM, etc.) is unimportant
- Each of these affiliations must be identified by an address and separated by a semicolon (;)
- The AP-HP institution must feature under the acronym "**AP-HP**" first in the address, specifically followed by: **AP-HP**, hospital, department, city, postcode, France.

16.2 Mention of the sponsor AP-HP (DRCI) in the acknowledgements of the text

The sponsor was Assistance Publique – Hôpitaux de Paris (Délégation à la Recherche Clinique et à l'Innovation).

16.3 Mention of the financial backer in the acknowledgements of the text

The study is funded by:
A grant from Programme Hospitalier de Recherche Clinique – PHRC-I 2025 (French Ministry of Health) "
A grant from "Institut de la leucémie" research program

This study has been registered on the website http://clinicaltrials.gov/ under number *(add the registration number when the study is registered)*.

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18 LIST OF ADDENDA

18.1 List of investigators

18.2 Serious Adverse Events notification form

18.3 Pregnancy notification form

18.4 Therapy-related malignancy notification form

18.5 SmPC or Investigator's Brochure

18.6 Recommended post-induction therapy

18.7 Inhibitors, substrates and inducers of CYP3A4

18.8 Statistical article (Andrillon, Chevret, Lee, Briard, 2020)

18.1 List of investigators

Centre investigateur coordonnateur							
N	Nom Investigateur coordonnateur	Prénom Investigateur coordonnateur	Courriel Investigateur coordonnateur	Etablissement de santé coordonnateur E_coordonnateur	Site d'exercice, GH coordonnateur	Ville	Pays
1	FODIL	Sofiane	sofiane.fodil@aphp.fr	AP-HP	AP-HP - Hôpital Saint Louis	Paris	FRANCE

Centre(s) co-investigateur(s)			Recrutants des participants : 5				
N+1	Nom co-investigateur(s)	Prénom co-investigateur(s)	Courriel co-investigateur(s)	Etablissement(s) de santé co-investigateur(s)	Site d'exercice, GH co-investigateur(s)	Ville	Pays
2	DE BOTTON	Stéphane	Stephane.DEBOTTON@gustaveroussy.fr	Gustave Roussy	Gustave Roussy	Villejuif	FRANCE
3	BRAUN	Thorsten	thorsten.braun@aphp.fr	AP-HP	Hopital Avicenne	Bobigny	FRANCE
4	BIRSEN	Rudy	rudy.birsen@aphp.fr	AP-HP	Hopital Cochin	Paris	FRANCE
5	CLUZEAU	Thomas	cluzeau.t@chu-nice.fr	CHU de Nice	Hopital de l'Archet	Nice	FRANCE
6	GARCIAZ	Sylvain	garciazs@ipc.unicancer.fr	Institut Paoli-Calmettes	Institut Paoli-Calmettes	Marseille	FRANCE

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18.2 Serious Adverse Events notification form

19 **Direction de la Recherche Clinique de l'Innovation, des relations avec les universités et les organismes de la Recherche (DRCI)**	ASSISTANCE PUBLIQUE HÔPITAUX DE PARIS	**Document type** **Réf. du doctype : QUA-DTYP-0235** **Date d'application du doctype : 21/06/2024**
	Formulaire de notification d'un Evènement Indésirable Grave (EIG) survenant au cours d'un essai clinique portant sur un Médicament ou produit assimilé	Réf. : REC-DTYP-0192 Version : 11 Date d'application : 06/05/2025 Emetteur : Valentin HOSANSKY

Dès la prise de connaissance de l'EIG par l'investigateur, ce formulaire doit être dûment complété (4 pages), daté et signé et retourné sans délai au secteur Vigilance de la DRCI par mail (eig-vigilance.drc@aphp.fr)

Il est possible de transmettre les formulaires de notification d'EIG au secteur Vigilance par télécopie au +33 (0)1 44 84 17 99 uniquement en cas de tentative infructueuse d'envoi par mail afin d'éviter les doublons.

Notification initiale ☐ **Suivi d'EIG** ☐ **N° du suivi** |__|__|

1. Identification de la recherche

Acronyme : ACTIVATE

Code de la Recherche : APHP251246

Date de notification : |__|__| |__|__| |2|0|__|__|
jj mm aaaa

Date de prise de connaissance de l'EIG par l'investigateur : |__|__| |__|__| |2|0|__|__|
jj mm aaaa

Titre complet de la Recherche :
Open label multi-center Phase 1/2 Clinical Trial Assessing safety, tolerability and efficacy of the combination of Actinomycin D and venetoclax in Relapsed/Refractory (R/R) Acute Myeloid Leukemia (AML)

Risque : ☐ A ☐ B ☐ C ☒ D

Plan expérimental :
☒ Essai non comparatif
☐ Essai comparatif ☐ Double aveugle ☐ Simple aveugle ☐ Ouvert
☐ Randomisé ☐ Non randomisé

2. Identification du centre investigateur

Nom de l'établissement : .....
Ville et code postal : .....
Service : .....

Investigateur (nom/prénom) : .....
Tél : ..... Fax : .....
Mail : .....

3. Identification et antécédents de la personne se prêtant à la recherche

Référence de la personne : |__|__|__| - |__|__|__|__|
n°centre – n° ordre de sélection

Sexe : ☐ M ☐ F

Poids : |__|__|__| kg
Taille : |__|__|__| cm

Date de naissance :
|__|__| |__|__|__|__|
mm aaaa

Age : |__|__|__|ans

Date de signature du consentement : |__|__| |__|__| |2|0|__|__|
jj mm aaaa

Date d'inclusion : |__|__| |__|__| |2|0|__|__|
jj mm aaaa

Antécédents médicaux-chirurgicaux/familiaux pertinents pour l'évaluation du cas (joindre un CRH anonymisé le cas échéant) :
.....
.....
.....
.....
.....
.....
.....

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4. Médicament(s) <u>expérimental(aux)</u> (ME) ou produit(s) assimilé(s) [préciser le(s)quel(s)] avant la survenue de l'EIG
(barrer l'encadré si traitement non débuté)

Nom commercial (de préférence) ou Dénomination Commune Internationale	Voie[1]	Posologie (préciser l'unité ex : mg/j)	Date de début *(jj/mm/aaaa)*	En cours[2]	Date de fin *(jj/mm/aaaa)*
ACTINOMYCIN D	.....	.....	|__|__| |__|__| |2|0|__|__| ☐ Non administré	☐	|__|__| |__|__| |2|0|__|__|
VENETOCLAX	.....	.....	|__|__| |__|__| |2|0|__|__| ☐ Non administré	☐	|__|__| |__|__| |2|0|__|__|

5. Médicament(s) <u>auxiliaire(s) nécessaire(s) à la réalisation de la recherche : si applicable</u> [préciser le(s)quel(s)] avant la survenue de l'EIG ***(barrer l'encadré si traitement non débuté)***

Nom commercial (de préférence) ou Dénomination Commune Internationale	Voie[1]	Posologie (préciser l'unité ex : mg/j)	Date de début *(jj/mm/aaaa)*	En cours[2]	Date de fin *(jj/mm/aaaa)*
Posaconazole	.....	.....	|__|__| |__|__| |2|0|__|__| ☐ Non administré	☐	|__|__| |__|__| |2|0|__|__|
....	.....	.....	|__|__| |__|__| |2|0|__|__| ☐ Non administré	☐	|__|__| |__|__| |2|0|__|__|

6. Procédures et actes ajoutés par la recherche ***(ex. : biopsies, IRM …)*** ***(barrer l'encadré si procédures et actes non réalisés)***	Date de réalisation *(jj/mm/aaaa)*	Chronologie	
		Avant la survenue de l'EIG	Après la survenue de l'EIG
Blood sample	|__|__| |__|__| |2|0|__|__| ☐ Non réalisé	☐	☐
Bone marrow aspiration	|__|__| |__|__| |2|0|__|__| ☐ Non réalisé	☐	☐
....	|__|__| |__|__| |2|0|__|__| ☐ Non réalisé	☐	☐

7. Médicament(s) <u>concomitant(s)</u> au moment de l'EIG, à l'exclusion de ceux utilisés pour traiter l'événement indésirable et ceux considérés comme médicaments auxiliaires ***(compléter le tableau ci-après et si nécessaire l'annexe relative aux médicaments concomitants ou barrer l'encadré si non applicable)***

➪ **Annexe jointe au présent formulaire :** ☐ Oui ☐ Non

Nom commercial (de préférence) ou Dénomination Commune Internationale	Voie[1]	Posologie (préciser l'unité ex : mg/j)	Dates d'administration *(du jj/mm/aa au jj/mm/aa)*	En cours[2]	Indication	***Action prise*** *0 : poursuite sans modification de la posologie 1 : arrêt 2 : diminution de la posologie 3 : augmentation de la posologie 4 : ne sais pas*	***Causalité de l'EIG*** *0 : non lié au médicament 1 : lié au médicament*
			du |__|__| |__|__| |__|__| au |__|__| |__|__| |__|__|	☐			
			du |__|__| |__|__| |__|__| au |__|__| |__|__| |__|__|	☐			

(1) Voie d'administration : VO=voie orale ; IM=Intramusculaire ; IV=intraveineuse ; SC=sous-cutanée ou autre (à préciser) (2) En cours au moment de la survenue de l'EIG

8. Evènement indésirable grave [EIG]

<u>Diagnostic</u> : ☐ Définitif ☐ Provisoire
.....
.....
.....
.....

<u>Date de survenue des premiers symptômes</u> : \|__\|__\| \|__\|__\| \|2\|0\|__\|__\|
Préciser lesquels :
.....

<u>Organe(s) concerné(s)</u> :
.....
.....
.....
.....
.....
.....

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Date d'apparition de l'EIG :
|__|__| |__|__| |_2_|_0_|__|__|
jj mm aaaa

Heure de survenue : |__||__| hh |__||__| min

☐ *donnée manquante*

Délai entre la date de la dernière administration du ME/produit assimilé ou la date de procédure/acte ajouté par la recherche et la date de survenue de l'EIG :
|__||__| / |__||__| |__||__|
jj hh min

Critères de gravité :

☐ Nécessite ou prolonge l'hospitalisation :
du |__|__| |__|__| |_2_|_0_|__|__|
au |__|__| |__|__| |_2_|_0_|__|__| ☐ *en cours*

☐ Décès
☐ Mise en jeu du pronostic vital
☐ Incapacité ou handicap important ou durable
☐ Anomalie ou malformation congénitale
☐ Autre(s) critère(s) médicalement significatif(s), préciser :
.....
...
.....
...

L'évènement a-t-il conduit à :

☐ aucune mesure prise concernant le ME
☐ diminution de la posologie du ME, préciser la nouvelle posologie : .....
date de début de la nouvelle posologie : |__|__| |__|__| |_2_|_0_|__|__|
☐ augmentation de la posologie du ME, préciser la nouvelle posologie : .....
date de début de la nouvelle posologie : |__|__| |__|__| |_2_|_0_|__|__|
☐ arrêt définitif du(des) ME, préciser la date : |__|__| |__|__| |_2_|_0_|__|__|
☐ arrêt transitoire du(des) ME, préciser la date : |__|__| |__|__| |_2_|_0_|__|__|
date de reprise : |__|__| |__|__| |_2_|_0_|__|__|
☐ ne sais pas

Récidive de l'EIG après ré-administration : ○ Non ○ Oui Date : |__|__| |__|__| |_2_|_0_|__|__|
○ Non applicable

Des mesures symptomatiques ont-elles été prises ?

☐ Non ☐ Oui Date : |__|__| |__|__| |_2_|_0_|__|__| Préciser : .....
.....

L'évènement fait-il suite à :

- une erreur médicamenteuse ?	☐ Non	☐ Oui	Date : |__|__| |__|__| |_2_|_0_|__|__|
- un surdosage ?	☐ Non	☐ Oui	Date : |__|__| |__|__| |_2_|_0_|__|__|
- un mésusage ?	☐ Non	☐ Oui	Date : |__|__| |__|__| |_2_|_0_|__|__|
- autre (préciser) :	☐ Non	☐ Oui	Date : |__|__| |__|__| |_2_|_0_|__|__|

Degré de sévérité :

Selon la classification CTCAE *:*
☐ Grade 1 ☐ Grade 2 ☐ Grade 3 ☐ Grade 4 ☐ Grade 5

Evolution de l'événement

☐ **Décès*** ☐ Evolution inconnue ☐ **Sujet non encore rétabli**, préciser :
○ Etat stable ○ Amélioration ○ Aggravation

Date : |__|__| |__|__| |_2_|_0_|__|__|
jj mm aaaa

**(à cocher uniquement si le décès est en lien avec cet/ces EIG)*

☐ **Résolu sans séquelles :**
Date : |__|__| |__|__| |_2_|_0_|__|__|
jj mm aaaa
|__|__| |__|__|
hh min

☐ **Résolu avec séquelles** :
Date de résolution de l'EIG :
|__|__| |__|__| |_2_|_0_|__|__|
jj mm aaaa
|__|__| |__|__|
hh min
Préciser les séquelles :
.....

Date de résolution des séquelles :
|__|__| |__|__| |_2_|_0_|__|__|
jj mm aaaa
|__|__| |__|__|
hh min

Si séquelles non résolues, condition chronique ?
☐ Oui
☐ Non *(merci de transmettre un suivi lors de la résolution des séquelles)*

9. Autre(s) étiologie(s) envisagée(s)

☐ Non ☐ Oui Si oui, préciser : .....
.....
.....

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10. Examen(s) complémentaire(s) réalisé(s)

☐ Non ☐Oui Si oui, préciser date, nature et résultats : [joindre les bilans anonymisés]

……

……

……

……

…..

11. Selon l'investigateur, l'événement indésirable grave est (plusieurs cases possibles)

Lié à la recherche :

☐ **au(x) médicament(s)/produit(s) assimilé(s) de la recherche : le(s)quel(s) ?**

Lequel : ACTINOMYCIN D …… ☐ Lié ☐ Non lié

Lequel : VENETOCLAX …… ☐ Lié ☐ Non lié

☐ **au(x) médicament(s) auxiliaire(s) nécessaire(s) à la réalisation de la recherche : Le(s)quel(s) ?**

Lequel : Posaconazole …… ☐ Lié ☐ Non lié

Lequel : …… ☐ Lié ☐ Non lié

☐ **à une interaction entre le(les) médicament(s) expérimental(aux) et le(les) médicament(s) auxiliaire(s) nécessaire(s) à la réalisation de la recherche ?**

☐ **Lié, préciser les médicaments suspectés pour l'interaction : ……**

☐ **Non lié**

☐ **à la (aux) procédure(s)/acte(s) de la recherche : la/le(s)quel(les) ?**

La/lequel(le) : Blood sample ☐ Lié ☐ Non lié

La/lequel(le) : Bone marrow aspiration ☐ Lié ☐ Non lié

Non lié à la recherche :

☐ **à la progression de la maladie faisant l'objet de la recherche :** Acute Myeloid Leukemia

☐ **à un (ou plusieurs) médicament(s) concomitant(s) administré(s), le(s)quel(s) :**

……

☐ **à une maladie intercurrente, laquelle :**

……

☐ **autre, préciser :**

……

Notificateur	**Investigateur**	Tampon du service :
Nom et fonction :	Nom :	
Signature :	Date et Signature :	

18.3 Pregnancy notification form

19 **Direction de la Recherche Clinique et de l'Innovation (DRCI)**	ASSISTANCE PUBLIQUE HÔPITAUX DE PARIS	**Document type** **Réf. du doctype : QUA-DTYP-0235** **Date d'application du doctype : 20/08/2020**
	Formulaire de notification et suivi d'une grossesse apparue au cours d'un essai clinique portant sur un Médicament ou produit assimilé	Réf. : REC-DTYP-0185 Version : 08 Date d'application : 18/06/2024 Emetteur : Sandrine COUE

Ce formulaire doit être dûment complété (3 pages), daté et signé, et retourné sans délai au secteur Vigilance de la DRCI par mail (eig-vigilance.drc@aphp.fr)

Il est possible de transmettre les formulaires de notification de grossesse au secteur Vigilance par télécopie au +33 (0)1 44 84 17 99 uniquement en cas de tentative infructueuse d'envoi par mail afin d'éviter les doublons.

1. Identification de la recherche **Notification initiale ☐** **Suivi de notification ☐ N° du suivi |__|__|**

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Acronyme : **ACTIVATE** Code de la recherche : APHP251246 Risque : D	**Date de notification :** |__|__| |__|__| |2|0|__|__| jj mm aaaa **Date de prise de connaissance de la grossesse par l'investigateur :** |__|__| |__|__| |2|0|__|__| jj mm aaaa

Titre complet de la Recherche : Open label multi-center Phase 1/2 Clinical Trial Assessing safety, tolerability and efficacy of the combination of Actinomycin D and venetoclax in Relapsed/Refractory (R/R) Acute Myeloid Leukemia (AML)

2. Identification du centre investigateur

Nom de l'établissement :	Investigateur (nom/prénom) : .....
Ville et code postal :	Tél : Fax :
Service :	E-mail :

3. Identification de la personne présentant une grossesse

Référence de la personne : |__|__|__| - |__|__|__|__|
n°centre – n° ordre de sélection –

Date de naissance : |__|__| |___|___|__|__| (mm-yyyy)
Date d'inclusion : |__|__| |__|__| |2|0|__|__|
Date de randomisation : |__|__| |__|__| |2|0|__|__|
Poids : |__|__|__| kg
Taille : |__|__|__| cm

Cas particulier d'une exposition paternelle : ☐ Oui ☐ Non

Référence de la personne : |__|__|__| - |__|__|__|__|
n°centre – n° ordre de sélection –

Date de naissance : |__|__| |___|___|__|__| (mm-yyyy)
Date d'inclusion : |__|__| |__|__| |2|0|__|__|
Date de randomisation : |__|__| |__|__| |2|0|__|__|

Expositions au cours de la grossesse :

Tabac :	☐ non	☐ oui (préciser nombre de paquets/année) :	☐ arrêt (préciser date) :	☐ poursuite
Alcool :	☐ non	☐ oui (préciser unités OH) :	☐ arrêt (préciser date) :	☐ poursuite
Drogue :	☐ non	☐ oui (préciser substance) :	☐ arrêt (préciser date) :	☐ poursuite

Autre (préciser) :

4. Antécédents maternels

Médicaux :	**Chirurgicaux :**

Obstétricaux : |__|__| geste |__|__| pare

Préciser si fausse couche, grossesse extra-utérine, interruption de grossesse (médicale ou volontaire), mort *in utero*, malformation congénitale, pathologie congénitale/néonatale non malformative, … (*nombre, date et nature/raison si applicable)*.

5. Médicament(s) expérimental (aux) administré(s) ou non pendant la grossesse ou s'il s'agit une exposition paternelle

Nom commercial (de préférence) ou Dénomination Commune Internationale	Date de première administration Ou non administré	Date de dernière administration Ou en cours	Voie d'administration(1)	Posologie / 24h
ACTINOMYCIN D	|__|__| |__|__| |2|0|__|__| ☐ Non administré	|__|__| |__|__| |2|0|__|__| ☐ En cours		
VENETOCLAX	|__|__| |__|__| |2|0|__|__| ☐ Non administré	|__|__| |__|__| |2|0|__|__| ☐ En cours		

(1) Voie d'administration : VO=voie orale ; IM=Intramusculaire ; IV=intraveineuse ; SC=sous-cutanée ou autre (à préciser)

6 . Médicament(s) auxiliaire(s) nécessaire(s) à la réalisation de la recherche administré(s) ou non pendant la grossesse ou s'il s'agit une exposition paternelle : si applicable

Nom commercial (de préférence) ou Dénomination Commune Internationale	Date de première administration Ou non administré	Date de dernière administration Ou en cours	Voie d'administration(1)	Posologie / 24h
Posaconazole	|__|__| |__|__| |2|0|__|__| ☐ Non administré	|__|__| |__|__| |2|0|__|__| ☐ En cours		
	|__|__| |__|__| |2|0|__|__| ☐ Non administré	|__|__| |__|__| |2|0|__|__| ☐ En cours		

7. Procédures et actes ajoutés par la recherche ***(Barrez l'encadré si procédures et actes non réalisés)***	Date de réalisation *(jj/mm/aaaa)*	Chronologie	
		Avant la grossesse	Au cours de la grossesse
Blood sample	|__|__| |__|__| |2|0|__|__| ☐ Non réalisé		
Bone marrow aspiration	|__|__| |__|__| |2|0|__|__| ☐ Non réalisé		

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	|__|__| |__|__| |_2_|_0_|__|__| ☐ Non réalisé		

8. Médicament(s) concomitants administré(s) dans le cadre du soin

(Cf. annexe « Liste relative aux médicaments concomitants » complétée : ☐ ***Oui*** ☐ ***Non applicable)***

Nom commercial (de préférence) ou Dénomination Commune Internationale	Date de première administration	Date de dernière administration Ou en cours	Voie d'administration(1)	Posologie / 24h
	|__|__| |__|__| |_2_|_0_|__|__|	|__|__| |__|__| |_2_|_0_|__|__| ☐ En cours		
	|__|__| |__|__| |_2_|_0_|__|__|	|__|__| |__|__| |_2_|_0_|__|__| ☐ En cours		
	|__|__| |__|__| |_2_|_0_|__|__|	|__|__| |__|__| |_2_|_0_|__|__| ☐ En cours		

(1) Voie d'administration : VO=voie orale ; IM=Intramusculaire ; IV=intraveineuse ; SC=sous-cutanée ou autre (à préciser)

9. Suivi de la grossesse

☐ Echographiques. Date(s) et résultats à préciser *(joindre les CR anonymisés)* :

☐ Autres examens. Date(s) et résultats à préciser *(joindre les CR anonymisés)* :

10. Grossesse en cours ☐ (envoyer par e-mail un nouveau formulaire complété à l'issue de la grossesse pour le suivi de la notification initiale)

ou issue de la grossesse ☐ (compléter ci-dessous)

Date : |__|__| |__|__| |_2_|_0_|__|__| Terme : |__|__| SA |__|__| J

☐ Fausse couche
→ Examen anatomo-pathologique disponible : ☐ Non ☐ Oui, précisez le résultat :

☐ Grossesse extra-utérine
→ Examen anatomo-pathologique disponible : ☐ Non ☐ Oui, précisez le résultat :

☐ Interruption de grossesse → Raison :
→ Examen anatomo-pathologique disponible : ☐ Non ☐ Oui, précisez le résultat :

☐ Accouchement : ☐ Spontané ☐ Provoqué ☐ Voie basse ☐ Césarienne

Naissance multiple : ☐ Non ☐ Oui, précisez le nombre :
Souffrance fœtale : ☐ Non ☐ Oui, précisez :
Mort-né : ☐ Non ☐ Oui, précisez :
Placenta normal : ☐ Oui ☐ Non, précisez :
Liquide amniotique : ☐ Clair ☐ Autre, précisez :
Anesthésie : ☐ Générale ☐ Péridurale ☐ Rachianesthésie ☐ Aucune

11. Nouveau-né (Si naissance multiple, compléter les parties 1, 2, 3, 9 et 10 d'un nouveau formulaire et l'envoyer par e-mail)

Sexe : ☐ Masculin ☐ Féminin

Poids : |__|__|__|__| grammes Taille : |__|__|__| cm Périmètre crânien : |__|__|__| cm

APGAR : 1 minute : _____ 5 minutes : _____ 10 minutes : _____

Malformation(s) congénitale(s) : ☐ Non ☐ Oui, précisez :

Pathologie(s) congénitale(s)/néonatale(s) non malformative(s) : ☐ Non ☐ Oui, précisez :

Le nouveau-né a-t-il bénéficié d'un suivi particulier à la naissance : ☐ Non ☐ Oui, précisez : ☐ Non applicable

Notificateur	**Investigateur**	Tampon du service :
Nom et fonction : Signature :	Nom : Date et signature :	

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18.4 Therapy-related malignancy notification form

19 **Direction de la Recherche Clinique de l'Innovation, des relations avec les universités et les organismes de la Recherche (DRCI)**	ASSISTANCE PUBLIQUE HÔPITAUX DE PARIS	**Document type** **Réf. du doctype : QUA-DTYP-0235** **Date d'application du doctype : 21/06/2024**
	Formulaire de notification des cancers secondaires/myélodysplasies survenant au cours d'un essai clinique portant sur un Médicament ou produit assimilé	Réf. : REC-DTYP-0191 Version : 08 Date d'application : 06/05/2025 Emetteur : Valentin HOSANSKY

Ce formulaire doit être dûment complété (5 pages), daté et signé, et retourné sans délai au secteur Vigilance de la DRCI par mail (eig-vigilance.drc@aphp.fr)

Il est possible de transmettre les formulaires de notification de cancers secondaires/myélodysplasies au secteur Vigilance par télécopie au +33 (0)1 44 84 17 99 uniquement en cas de tentative infructueuse d'envoi par mail afin d'éviter les doublons.

Notification initiale ☐ **Suivi de notification** ☐ **N° du suivi** |__|__|

1. Identification de la recherche

Acronyme : ACTIVATE

Code de la Recherche : APHP251246

Titre complet de la Recherche :
Open label multi-center Phase 1/2 Clinical Trial Assessing safety, tolerability and efficacy of the combination of Actinomycin D and venetoclax in Relapsed/Refractory (R/R) Acute Myeloid Leukemia (AML)

Date de notification : |__|__| |__|__| |2|0|__|__|
j j m m a a a a

Date de prise de connaissance du des cancer secondaire/myélodysplasie par l'investigateur : |__|__| |__|__| |2|0|__|__|
j j m m a a a a

Risque : ☐ A ☐ B ☐ C ☒ D

Plan expérimental : ☒ Essai non comparatif
☐ Essai comparatif ☐ Double aveugle ☐ Simple aveugle ☐ Ouvert
☐ Randomisé ☐ Non randomisé

2. Identification du centre investigateur

Nom de l'établissement : .....
Ville et code postal : .....
Service : .....

Investigateur (nom/prénom) : .....
.....
Tél : ..... Fax : .....
Mail : .....

3. Identification et antécédents de la personne se prêtant à la recherche

Référence de la personne : |__|__|__| - |__|__|__|__|
n°centre – n° ordre de sélection

Sexe : ☐ M ☐ F

Poids : |__|__|__| kg
Taille : |__|__|__| cm

Date de naissance :
|__|__| |__|__|__|__|
m m a a a a

Age : |__|__|__|ans

Date de signature du consentement : |__|__| |__|__| |2|0|__|__|
j j m m a a a a

Date d'inclusion : |__|__| |__|__| |2|0|__|__|
jj mm aaaa

Antécédents médicaux-chirurgicaux/familiaux pertinents pour l'évaluation du cas (joindre un CRH anonymisé le cas échéant) :
.....
.....
.....
.....
.....
.....
.....
.....
.....
.....
.....
....

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4. Diagnostic du cancer secondaire/de la myélodysplasie	
4.1 Diagnostic clinique :	
Date du diagnostic : |__|__| |__|__| |_2_|_0_|__|__| j j m m a a a a	Diagnostic final retenu : ……
Confirmation histologique : ☐ **Non** ☐ **Oui**	
Confirmation cytologique : ☐ **Non** ☐ **Oui**	
4.2 Grade : (précisez l'échelle de classification ex : TNM)	☐ Grade 0 ☐ Grade I ☐ Grade II ☐ Grade III ☐ Grade IV ……
4.3 Grade histologique	☐ Grade 0 ☐ Grade I ☐ Grade II ☐ Grade III ☐ Grade IV
4.4 Si autre classification, précisez :	
4.5 Antécédents médicaux pertinents : ☐ Non ☐ Oui, précisez :	

5. Médicaments/procédures de la recherche

5.1 Médicament(s) <u>expérimental(aux)</u> (ME) ou produit(s) assimilé(s) [préciser le(s)quel(s)] avant la survenue de l'EIG
(barrer l'encadré si traitement non débuté)

Nom commercial (de préférence) ou Dénomination Commune Internationale	Voie(1)	Posologie (préciser l'unité ex : mg/j)	Date de début *(jj/mm/aaaa)*	En cours(2)	Date de fin *(jj/mm/aaaa)*
ACTINOMYCIN D	……	……	|__|__| |__|__| |_2_|_0_|__|__| ☐ Non administré	☐	|__|__| |__|__| |_2_|_0_|__|__|
VENETOCLAX	……	……	|__|__| |__|__| |_2_|_0_|__|__| ☐ Non administré	☐	|__|__| |__|__| |_2_|_0_|__|__|

5.2 Médicament(s) <u>auxiliaire(s) nécessaire(s) à la réalisation de la recherche : si applicable</u> [préciser le(s)quel(s)] avant la survenue de l'EIG
(barrer l'encadré si traitement non débuté)

Nom commercial (de préférence) ou Dénomination Commune Internationale	Voie(1)	Posologie (préciser l'unité ex : mg/j)	Date de début *(jj/mm/aaaa)*	En cours(2)	Date de fin *(jj/mm/aaaa)*
Posaconazole	……	……	|__|__| |__|__| |_2_|_0_|__|__| ☐ Non administré	☐	|__|__| |__|__| |_2_|_0_|__|__|
……	……	……	|__|__| |__|__| |_2_|_0_|__|__| ☐ Non administré	☐	|__|__| |__|__| |_2_|_0_|__|__|

5.3 Procédures et actes ajoutés par la recherche ***(ex. : biopsies, IRM …)*** ***(barrer l'encadré si procédures et actes non réalisés)***	Date de réalisation *(jj/mm/aaaa)*	Chronologie	
		Avant la survenue de l'EIG	Après la survenue de l'EIG
Blood sample	|__|__| |__|__| |_2_|_0_|__|__| ☐ Non réalisé	☐	☐
Bone marrow aspiration	|__|__| |__|__| |_2_|_0_|__|__| ☐ Non réalisé	☐	☐

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6. Précision de l'imputabilité de l'investigateur

6.1 Selon l'investigateur, l'événement indésirable grave (cancer secondaire/myélodysplasie) est (plusieurs cases possibles)

Lié à la recherche :

☐ **au(x) médicament(s)/produit(s) assimilé(s) de la recherche : le(s)quel(s) ?**

Lequel : … ACTINOMYCIN D …… ☐ **Lié** ☐ **Non lié**

Lequel : … VENETOCLAX …… ☐ **Lié** ☐ **Non lié**

☐ **au(x) médicament(s) auxiliaire(s) nécessaire(s) à la réalisation de la recherche : Le(s)quel(s) ?**

Lequel : … Posaconazole …… ☐ **Lié** ☐ **Non lié**

Lequel : …… ☐ **Lié** ☐ **Non lié**

☐ **à une interaction entre le(les) médicament(s) expérimental(aux) et le(les) médicament(s) auxiliaire(s) nécessaire(s) à la réalisation de la recherche ?**

☐ **Lié, préciser les médicaments suspectés pour l'interaction : …….**

☐ **Non lié**

☐ **à la (aux) procédure(s)/acte(s) de la recherche : la/le(s)quel(les) ?**

La/lequel(le) : Blood sample ☐ **Lié** ☐ **Non lié**

La/lequel(le) : Bone marrow aspiration ☐ **Lié** ☐ **Non lié**

Non lié à a recherche :

☐ **à la progression de la maladie faisant l'objet de la recherche :** Acute Myeloid Leukemia

☐ **à un (ou plusieurs) médicament(s) concomitant(s) administré(s), le(s)quel(s) : ……**

☐ **à une maladie intercurrente, laquelle : ……**

☐ **autre, préciser : ……**

6.2 La survenue de cet EIG est-elle susceptible d'être liée à un manque d'efficacité du ME ?	☐ **Non** ☐ **Oui**

7. Détails de la chimiothérapie administrée pour traiter la pathologie initiale (phase)

☐ Induction |__|__| |__|__| |_2_|_0_|__|__| j j m m a a a a	☐ Consolidation |__|__| |__|__| |_2_|_0_|__|__| j j m m a a a a
☐ Post greffe : renseignez la partie 6.2	☐ Maintenance |__|__| |__|__| |_2_|_0_|__|__| j j m m a a a a
☐ Autre :	☐ Interphase

7.1 Médicament(s) ou produit(s) assimilé(s) de chimiothérapie anticancéreuse ou de thérapie ciblée avant la survenue du cancer secondaire/de la myélodysplasie *(barrez l'encadré si aucun traitement débuté) :*

Nom commercial ou Dénomination Commune International	Date de première administration Ou non administré	Date de dernière administration Ou en cours (2)	Voie d'administration(1)	Posologie / 24h	Lien de causalité avec l'EIG *(Relation selon méthode EVCTM)*
	|__|__| |__|__||_2_|_0_|__|__| j j m m a a a a ☐ Non administré	|__|__| |__|__||_2_|_0_|__|__| j j m m a a a a ☐ En cours			☐ non lié ☐ lié
	|__|__| |__|__||_2_|_0_|__|__| j j m m a a a a ☐ Non administré	|__|__| |__|__||_2_|_0_|__|__| j j m m a a a a ☐ En cours			☐ non lié ☐ lié

(1) Voie d'administration : VO=voie orale ; IM=Intramusculaire ; IV=intraveineuse ; SC=sous-cutanée ou autre (à préciser)

(2) En cours au moment de la survenue de l'EIG

7.2 Greffe de cellules souches hématopoïétiques (CSH) pour le traitement de la pathologie initiale :

☐ Non ☐ Oui, précisez ci-dessous :

Date de la greffe : le |__|__| |__|__| |_2_|_0_|__|__| j j m m a a a a ☐ autogreffe ☐ allogreffe	Si allogreffe : Donneur : ☐ apparenté ☐ fichier volontaires / banque
Origine CSH : ☐ CSP ☐ Moelle osseuse ☐ Sang de cordon	
Date de sortie d'aplasie : |__|__| |__|__| |_2_|_0_|__|__| j j m m a a a a	

7.3 Traitements de conditionnement de la greffe (immunosuppresseurs, irradiation corporelle, etc.) :

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☐ Non applicable ☐ Applicable, précisez ci-dessous le schéma thérapeutique :

Nom commercial ou Dénomination Commune Internationale	Date de première administration	Date de dernière administration	Voie d'administration(1)	Posologie / 24h
	|__|__| |__|__| |_2_|_0_|__|__|	|__|__| |__|__| |_2_|_0_|__|__|		
	|__|__| |__|__| |_2_|_0_|__|__|	|__|__| |__|__| |_2_|_0_|__|__|		

8. Statut de la pathologie initiale à la date de survenue du cancer secondaire/de la myélodysplasie

(Joindre si possible les résultats du dernier myélogramme le cas échéant) **:**

☐ Rémission complète le \|__\|__\| \|__\|__\| \|_2_\|_0_\|__\|__\|
☐ Rémission avec séquelles le \|__\|__\| \|__\|__\| \|_2_\|_0_\|__\|__\|, précisez les séquelles :
☐ Rémission partielle le \|__\|__\| \|__\|__\| \|_2_\|_0_\|__\|__\|, précisez :
☐ Stable depuis le \|__\|__\| \|__\|__\| \|_2_\|_0_\|__\|__\|
☐ Maladie en progression
☐ Rechute depuis le\|__\|__\| \|__\|__\| \|_2_\|_0_\|__\|__\|

9. Traitement du cancer secondaire/de la myélodysplasie

9.1 Hospitalisation(s) :

Hospitalisation (1) du \|__\|__\| \|__\|__\| \|_2_\|_0_\|__\|__\| au \|__\|__\| \|__\|__\| \|_2_\|_0_\|__\|__\|
Hospitalisation (2) du \|__\|__\| \|__\|__\| \|_2_\|_0_\|__\|__\| au \|__\|__\| \|__\|__\| \|_2_\|_0_\|__\|__\|
Hospitalisation (3) du \|__\|__\| \|__\|__\| \|_2_\|_0_\|__\|__\| au \|__\|__\| \|__\|__\| \|_2_\|_0_\|__\|__\|

9.2 Intervention chirurgicale : ☐ Non ☐ Oui, précisez ci-dessous :

Type d'intervention chirurgicale :	Date de l'intervention chirurgicale : |__|__| |__|__| |_2_|_0_|__|__|

9.3 Chimiothérapie : ☐ Non ☐ Oui, précisez ci-dessous :

Précisez le schéma thérapeutique, date(s) de début, les posologies et dates de fin si applicable :

9.4 Radiothérapie : ☐ Non ☐ Oui, précisez ci-dessous :

Précisez le schéma thérapeutique et les doses :	Date de début : |__|__| |__|__| |_2_|_0_|__|__|	Date de fin : |__|__| |__|__| |_2_|_0_|__|__|

9.5 Traitement(s) adjuvant(s) : ☐ Non ☐ Oui, précisez ci-dessous :

9.6 Une greffe de CSH a été réalisée pour le traitement du cancer secondaire/de la myélodysplasie : ☐ Non ☐ Oui, précisez ci-dessous :

Date de la greffe : le |__|__| |__|__| |_2_|_0_|__|__| ☐ autogreffe ☐ allogreffe	Si allogreffe : Donneur : ☐apparenté ☐ fichier volontaires / banque

Origine CSH : ☐ CSP ☐ Moelle osseuse ☐ Sang de cordon

Date de sortie d'aplasie : \|__\|__\| \|__\|__\| \|_2_\|_0_\|__\|__\|

10. Evolution du cancer secondaire/de la myélodysplasie

10.1 Etat actuel (hors décès)

☐ Rémission complète le \|__\|__\| \|__\|__\| \|_2_\|_0_\|__\|__\|
☐ Rémission avec séquelles le \|__\|__\| \|__\|__\| \|_2_\|_0_\|__\|__\|, précisez les séquelles :
☐ Rémission partielle le \|__\|__\| \|__\|__\| \|_2_\|_0_\|__\|__\|, précisez :
☐ Stable depuis le \|__\|__\| \|__\|__\| \|_2_\|_0_\|__\|__\|
☐ Maladie en progression
☐ Rechute depuis le\|__\|__\| \|__\|__\| \|_2_\|_0_\|__\|__\|

10.2 Evolution fatale

Date du décès : \|__\|__\| \|__\|__\| \|_2_\|_0_\|__\|__\|

Autopsie effectuée : ☐ Non ☐ Oui (joindre le compte-rendu)

Veuillez spécifier la « cause du décès » rapportée dans le certificat de décès / le rapport d'autopsie :

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Notificateur	**Investigateur**	Tampon du service :
Nom et fonction : Signature :	Nom : Date et Signature :	

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18.5 SmPC or Investigator's Brochure

SmPC for Actinomycin D and Venetoclax are available on: http://agence-prd.ansm.sante.fr/php/ecodex/index.php

18.6 Recommended post-induction therapy

Patients deriving clinical benefit (defined as achievement of MLFS, CRi, CRh or CR) after 2 cycles may receive up to 10 consolidation cycles outside of the study, at the investigator's discretion, consisting of:

- Venetoclax daily days 1-7,
- And Actinomycin D by short IV infusion 12.5 µg/kg/d at day 3.

Patients with a clinical benefit eligible for allogeneic transplantation, at the investigator's discretion, may receive up to 6 cycles prior to transplantation, outside of the study.

18.7 Inhibitors, substrates and inducers of CYP3A4

Inhibiteurs	**Substrats**	**Inducteurs**
Clarithromycine	Rivaroxaban	Rifampicine
Érythromycine	Apixaban	Phénytoïne
Kétoconazole	Ticagrelor	Carbamazépine
Itraconazole	Atorvastatine	Phénobarbital
Voriconazole	Simvastatine	Millepertuis
Ritonavi	Ibrutinib	Efavirenz
Diltiazem		
Vérapamil		
Amiodarone		
Fluconazole		
Cimétidine		

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19 Résumé synoptique du protocole (sur 2 pages maximum, calibri 10)

Full Title: Open-label, Multicenter Phase 1 Clinical Trial Assessing Safety, Tolerability, and Efficacy of Actinomycin D and Venetoclax in Relapsed/Refractory (R/R) Acute Myeloid Leukemia (AML)

Acronym: ACTIVATE

Coordinating Investigator: Dr. Sofiane Fodil

Scientific Director: Prof. Raphaël Itzykson

Sponsor : Assistance Publique – Hôpitaux de Paris

Scientific Rationale

AML is a heterogeneous myeloid malignancy characterized by clonal expansion of hematopoietic progenitors, leading to bone marrow failure and poor survival outcomes. Despite advances in frontline treatment (e.g., 7+3 chemotherapy, HSCT), one-third of AML cases remain refractory, and half of the responders relapse. With limited targeted therapies beyond FLT3 and IDH1/2 inhibitors, novel strategies for R/R AML are urgently needed.

Rationale for Actinomycin D + Venetoclax Combination

Venetoclax, a BCL-2 inhibitor, disrupts apoptosis resistance but is counteracted by MCL-1 upregulation, a major resistance mechanism. Actinomycin D targets RNA Polymerase I, induces nucleolar stress, TP53 activation, and MCL-1 downregulation, thereby restoring Venetoclax sensitivity. Retrospective data report response rates from 26% to 75% in R/R AML patients treated with this combination, warranting prospective validation in a clinical setting.

Study Design & Population

Phase 1, Open-label, Multicenter Trial (6 sites in France), Adaptive Bayesian TITE-CRM Design

36 Patients with R/R AML post-intensive chemotherapy

Duration: 36 months (Enrollment: 18 months, Treatment: 6 months, Follow-up: 12 months)

Objectives & Endpoints

Primary Objective

Assess safety & dose-limiting toxicities (DLTs) to define the maximum tolerated dose (MTD)

DLT Probability ≤ 33% at the end of the first cycle (Days 28-42)

Expected ORR: ≥ 33% after 1-2 cycles

Primary Endpoints

DLT Assessment: Death, prolonged myelosuppression, severe non-hematologic toxicities

Safety Monitoring: lab assessments, adverse event tracking (NCI CTCAE v5.0)

Secondary Objectives & Endpoints

Safety Profile: TEAEs, SAEs, long-term adverse events

Efficacy Indicators: ORR, CR, CRi, MLFS, EFS, OS

Pharmacodynamics: Leukemia cell response biomarkers (PB/BM analysis)

Treatment Plan

Induction Phase (2 cycles):

Venetoclax (Days 1-7) + Actinomycin D (Days 3-5, escalating DL1-DL3)

Consolidation Phase (up to 10 cycles) (outside trial scope)

Allogeneic HSCT Consideration for eligible patients

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Eligibility Criteria

Inclusion

R/R AML post-intensive chemotherapy

Age ≥18 years, ECOG ≤2, WBC <20,000/mm³

No oncogenetic restrictions (subtype-independent efficacy)

Exclusion

APL, active CNS involvement, prior HSCT (<12 weeks)

Uncontrolled infections, concurrent malignancies requiring therapy

Pregnancy, breastfeeding

Potential Impact on Patient Care

This study could introduce a novel salvage regimen for R/R AML, offering higher response rates, prolonged survival, and reduced toxicity compared to conventional salvage therapies. If successful, these findings may lead to changes in treatment guidelines and establish Actinomycin D + Venetoclax as a viable alternative to intensive chemotherapy-based salvage regimens.

Safety & Risk Monitoring

Potential Risks: Myelosuppression, infections, TLS, GI toxicity

Mitigation Strategies: Hydration protocols, lab monitoring

Regular DSMB reviews (3-member board)

Predefined stopping criteria for safety concerns

Statistical Considerations

Bayesian TITE-CRM model

Sample size (36 patients) ensures safety & efficacy insights

Target ORR: ≥50%, MTD DLT ≤33%

Regulatory & Funding

Main Funding: PHRC-I (if obtained), IHU THEMA

EMA/ICH-GCP compliance

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